IWoAT 2026

Lecture Notes — Days 1–5

An Introduction to Stable Homotopy Theory
and Brave New Algebra

Based on lectures by

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A few passages headed *Understanding / Intuition* are my own annotations. The notes have not been reviewed or approved by the lecturers.

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Part I

Day 1: Abstract Homotopy Theory

Day 1 was background. We start with a category and a class of maps we want to treat as equivalences, and then look at three ways of working with the resulting homotopy theory: model categories, simplicial enrichment, and quasicategories. I state most results without their precise technical hypotheses; Hovey's book [Hov99] has them for the model-category part.

1 Weak Equivalences and Model Categories

1.1 Homotopical categories

Definition 1.1 (Homotopical category)**.** A *homotopical category* is a category $\mathcal{C}$ with a subcategory $W \subseteq \mathcal{C}$ of *weak equivalences* that contains all identities and has the two-out-of-six property: whenever

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

are composable and both gf and hg lie in W, then f, g, h, and hgf all lie in W.

Two-out-of-six implies two-out-of-three, but it still does not give closure under retracts.

Definition 1.2 (Homotopical functor)**.** A functor $F\colon \mathcal{C} \to \mathcal{D}$ between homotopical categories $(\mathcal{C}, W_{\mathcal{C}})$ and $(\mathcal{D}, W_{\mathcal{D}})$ is *homotopical* if it sends $W_{\mathcal{C}}$ into $W_{\mathcal{D}}$.

These are the functors that pass to the localizations for free. Most functors we care about are not homotopical, which is why derived functors are needed later.

1.2 The model-category axioms

Definition 1.3 (Model category)**.** A *model category* is a complete and cocomplete category $\mathcal{M}$ with three classes of morphisms,

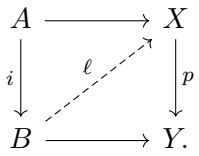
$$W \quad \text{(weak equivalences)}, \qquad C \quad \text{(cofibrations)}, \qquad F \quad \text{(fibrations)},$$

such that:

(MC1) W satisfies two-out-of-three.
(MC2) Each of W, C, and F is closed under retracts.
(MC3) Cofibrations have the left lifting property against acyclic fibrations, and acyclic cofibrations have it against fibrations. (*Acyclic* means "also a weak equivalence.")
(MC4) Every map factors as a cofibration followed by an acyclic fibration, and as an acyclic cofibration followed by a fibration.

In other words, $(C, F \cap W)$ and $(C \cap W, F)$ are weak factorization systems. We assume the factorizations can be chosen functorially, since we want replacement functors.

The lifting axiom is about commutative squares



We write $i \,\square\, p$ if a diagonal ℓ exists for every such square, so (MC3) reads

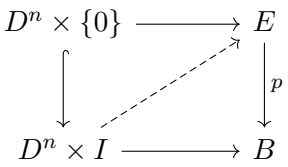
$$C \,\square\, (F \cap W), \qquad (C \cap W) \,\square\, F.$$

Neither lifts nor factorizations are unique, but any two choices differ by a weak equivalence, which is all we need.

Definition 1.4 (Fibrant and cofibrant objects)**.** Let $\varnothing$ and $*$ be the initial and terminal objects of $\mathcal{M}$. An object X is *cofibrant* if $\varnothing \to X$ is a cofibration, *fibrant* if $X \to *$ is a fibration, and *bifibrant* if it is both.

Example 1.5. The two examples to keep in mind are spaces and simplicial sets. In the Quillen model structure on topological spaces, the weak equivalences are the weak homotopy equivalences and the fibrations are the Serre fibrations. In the Kan–Quillen model structure on simplicial sets, the weak equivalences are the maps whose geometric realization is a weak homotopy equivalence, the cofibrations are the monomorphisms, and the fibrant objects are the Kan complexes.

Recall that $p\colon E \to B$ is a *Serre fibration* if it has the homotopy lifting property for disks, that is, the right lifting property in



for every $n \geq 0$.

Example 1.6 (The projective model structure on connective complexes)**.** Let R be a ring and $\mathbf{Ch}_{\geq 0}(R)$ the category of nonnegatively graded chain complexes of left R-modules. In the projective model structure the weak equivalences are the quasi-isomorphisms and the fibrations are the maps that are surjective in positive degrees. The cofibrations are whatever has the left lifting property against acyclic fibrations. The cofibrant complexes turn out to be the degreewise projective ones, so cofibrant replacement is a projective resolution.

Convention 1.7 (Arrow notation)**.** In diagrams, $\rightarrowtail$ is a cofibration, $\twoheadrightarrow$ is a fibration, and a $\sim$ on an arrow marks a weak equivalence. So an acyclic cofibration is drawn $\xhookrightarrow{\sim}$ and an acyclic fibration $\xrightarrow{\sim}\!\!\!\!\twoheadrightarrow$.

1.3 Cofibrant and fibrant replacement

Replacements come from applying (MC4) to the maps out of the initial object and into the terminal object. Factoring $\varnothing \to X$ as a cofibration followed by an acyclic fibration gives

$$\varnothing \hookrightarrow QX \xrightarrow[q_X]{\sim}\!\!\!\!\twoheadrightarrow X,$$

with QX cofibrant. Factoring $X \to *$ as an acyclic cofibration followed by a fibration gives

$$X \xhookrightarrow[\sim]{r_X} RX \twoheadrightarrow *,$$

with RX fibrant. With functorial factorizations, Q and R are the cofibrant and fibrant replacement functors. The directions are easy to mix up: the cofibrant replacement maps *to* X, and the fibrant replacement receives a map *from* X.

Doing both, $X \leftarrow QX \to RQX$, gives a bifibrant replacement. So every object is weakly equivalent to one in the full subcategory $\mathcal{M}^{\mathrm{cf}}$ of bifibrant objects.

1.4 Cylinder objects, path objects, and homotopy

A cylinder object for X is a factorization of the fold map,

$$X \amalg X \xhookrightarrow{(i_0,i_1)} \mathrm{Cyl}(X) \xrightarrow{\sim} X.$$

A *left homotopy* between $f, g\colon X \to Y$ is a map $H\colon \mathrm{Cyl}(X) \to Y$ that restricts to f and g along i_0 and i_1. Dually, a path object for Y is a factorization of the diagonal,

$$Y \xrightarrow{\sim} \mathrm{Path}(Y) \xrightarrow{(p_0,p_1)} Y \times Y,$$

and a *right homotopy* is a map $X \to \mathrm{Path}(Y)$ with endpoints f and g.

When X is cofibrant and Y is fibrant, left and right homotopy are the same equivalence relation, and homotopy classes of maps $X \to Y$ are the morphisms in the homotopy category. Without these two hypotheses the relations can differ and need not even be transitive.

2 The Homotopy Category and the Calculus of Fractions

2.1 Localization

Definition 2.1 (Localization and homotopy category)**.** Let $(\mathcal{C}, W)$ be a category with weak equivalences. Its *localization* is a functor

$$\gamma\colon \mathcal{C} \longrightarrow \mathcal{C}[W^{-1}]$$

that sends W to isomorphisms and is universal with this property: for every $\mathcal{D}$, precomposition with γ identifies functors $\mathcal{C}[W^{-1}] \to \mathcal{D}$ with functors $\mathcal{C} \to \mathcal{D}$ that invert W. For a model category we write

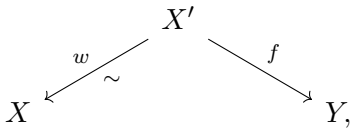
$$\mathrm{Ho}(\mathcal{M}) = \mathcal{M}[W^{-1}]$$

and call it the *homotopy category*.

Concretely, one adds a formal inverse for each weak equivalence, so a morphism in $\mathcal{C}[W^{-1}]$ is represented by a zigzag

$$X = X_0 \longleftarrow X_1 \longrightarrow X_2 \longleftarrow \cdots \longrightarrow X_n = Y$$

whose backward arrows are weak equivalences. This is hard to work with. If W satisfies an Ore-type condition, every zigzag can be shortened to a single roof



representing $\quad f\,w^{-1}$,

and two roofs are identified when they have a common refinement. This calculus of fractions is not available for a general $(\mathcal{C}, W)$. In a model category one gets something just as usable by replacing the source and the target:

Proposition 2.2 (Computing morphisms in $\mathrm{Ho}(\mathcal{M})$)**.** *For $X, Y \in \mathcal{M}$ there is a natural bijection*

$$\mathrm{Ho}(\mathcal{M})(X, Y) \cong \mathcal{M}(QX, RY)/\simeq,$$

where $\simeq$ is left (or right) homotopy. In particular $\mathrm{Ho}(\mathcal{M})$ *is equivalent to the category of bifibrant objects and homotopy classes of maps.*

Idea. The maps $QX \to X$ and $Y \to RY$ become isomorphisms in $\mathrm{Ho}(\mathcal{M})$, so we may assume X is cofibrant and Y fibrant. In that case the axioms let one turn any zigzag into an honest map, well defined up to homotopy. See [Hov99, Chapter 1]. □

3 Derived Functors

3.1 Total derived functors

Definition 3.1 (Total derived functors)**.** Suppose $F\colon \mathcal{M} \to \mathcal{N}$ preserves weak equivalences between cofibrant objects. Its *total left derived functor* is

$$\mathbf{L}F\colon \mathrm{Ho}(\mathcal{M}) \longrightarrow \mathrm{Ho}(\mathcal{N}), \qquad \mathbf{L}F(X) = F(QX).$$

If instead $G\colon \mathcal{M} \to \mathcal{N}$ preserves weak equivalences between fibrant objects, its *total right derived functor* is

$$\mathbf{R}G\colon \mathrm{Ho}(\mathcal{M}) \longrightarrow \mathrm{Ho}(\mathcal{N}), \qquad \mathbf{R}G(X) = G(RX).$$

Up to canonical isomorphism these do not depend on the choice of Q and R. The maps $QX \to X$ and $X \to RX$ give comparison transformations

$$(\mathbf{L}F)\gamma_{\mathcal{M}} \Longrightarrow \gamma_{\mathcal{N}} F, \qquad \gamma_{\mathcal{N}} G \Longrightarrow (\mathbf{R}G)\gamma_{\mathcal{M}},$$

which say that $\mathbf{L}F$ is the best approximation to F from the left by a functor on homotopy categories, and $\mathbf{R}G$ is the best one from the right.

3.2 Quillen adjunctions

Definition 3.2 (Quillen adjunction)**.** An adjunction

$$\mathcal{M} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{N} \qquad (F \dashv G)$$

is a *Quillen adjunction* if F preserves cofibrations and acyclic cofibrations, or equivalently G preserves fibrations and acyclic fibrations. It is a *Quillen equivalence* if, for cofibrant $X \in \mathcal{M}$ and fibrant $Y \in \mathcal{N}$, a map $FX \to Y$ is a weak equivalence exactly when its adjoint $X \to GY$ is.

The point of the definition is that it is easy to check and gives derived functors on both sides at once.

Proposition 3.3 (Derived adjunction)**.** *A Quillen adjunction $F \dashv G$ induces an adjunction*

$$\mathrm{Ho}(\mathcal{M}) \underset{\mathbf{R}G}{\overset{\mathbf{L}F}{\rightleftarrows}} \mathrm{Ho}(\mathcal{N}),$$

which is an equivalence of categories when $F \dashv G$ is a Quillen equivalence.

4 Diagram Categories and Homotopy (Co)limits

4.1 Model structures on diagram categories

For a small category I write $\mathcal{M}^I = \mathrm{Fun}(I, \mathcal{M})$. A map of diagrams is a weak equivalence if it is one at every object of I.

Definition 4.1 (Projective and injective diagram structures)**.** The *projective model structure* on $\mathcal{M}^I$ has objectwise weak equivalences and objectwise fibrations. The *injective model structure* has objectwise weak equivalences and objectwise cofibrations. In each case the third class is determined by lifting.

Neither structure exists in general, but both do when $\mathcal{M}$ is combinatorial. Reedy model structures are a third option when I is a Reedy category.

Understanding / Intuition

Being cofibrant in the projective structure is a condition on the whole diagram. A diagram whose entries are all cofibrant can still fail to be projectively cofibrant, because the structure maps matter too. The span $* \leftarrow A \rightarrow *$ in the next subsection is an example.

Colimit and limit are the two adjoints of the constant-diagram functor Δ:

$$\mathcal{M}^I \underset{\lim_I}{\overset{\mathrm{colim}_I}{\leftarrow\!\!\!

and its colimit is the pushout:

$$\begin{array}{ccc} A & \xrightarrow{g} & C \\ {\scriptstyle f}\downarrow & & \downarrow \\ B & \longrightarrow & B \amalg_A C. \end{array}$$

The homotopy pushout is the homotopy colimit of the span,

$$B \amalg^h_A C := \operatorname{hocolim}(B \leftarrow A \rightarrow C),$$

and it agrees with the ordinary pushout when the span is projectively cofibrant (all three objects cofibrant and both legs cofibrations). For spaces there is a model that needs no replacement, the double mapping cylinder

$$B \ \amalg_{A\times\{0\}} \ (A \times I) \ \amalg_{A\times\{1\}} \ C. \tag{4.3}$$

Collapsing the interval maps it to the ordinary pushout, and this collapse is a weak equivalence when, for instance, one of the two legs is a cofibration.

Example 4.3. For the span $* \leftarrow A \rightarrow *$ the ordinary pushout is a point, but the homotopy pushout is the unreduced suspension ΣA.

4.3 Pullbacks as homotopy limits

Dually, the limit of a cospan $B \xrightarrow{f} D \xleftarrow{g} C$ is the pullback

$$\begin{array}{ccc} B \times_D C & \longrightarrow & C \\ \downarrow & & {\scriptstyle g}\downarrow \\ B & \xrightarrow[f]{} & D, \end{array}$$

which for sets is

$$B \times_D C = \{(b, c) \mid f(b) = g(c)\}.$$

The homotopy pullback is the derived version,

$$B \times^h_D C := \operatorname{holim}(B \rightarrow D \leftarrow C).$$

For spaces, the standard model replaces the equation $f(b) = g(c)$ by a path between the two points:

$$\big\{(b, \gamma, c) \mid \gamma\colon I \rightarrow D,\ \gamma(0) = f(b),\ \gamma(1) = g(c)\big\}. \tag{4.4}$$

The ordinary pullback already computes the homotopy pullback if the cospan is fibrant, and in a right proper model category it is enough that one of the two legs is a fibration.

5 Simplicial Sets and Simplicial Enrichment

5.1 Simplicial sets

Let Δ be the simplex category: its objects are the finite ordered sets

$$[n] = \{0 < 1 < \cdots < n\}, \qquad n \geq 0,$$

and its morphisms are the order-preserving maps.

Definition 5.1 (Simplicial set)**.** A *simplicial set* is a functor

$$X\colon \Delta^{\mathrm{op}} \longrightarrow \mathbf{Set}.$$

The elements of $X_n = X([n])$ are the n-simplices. The injections and surjections between adjacent $[n]$ induce face maps $d_i\colon X_n \to X_{n-1}$ and degeneracy maps $s_i\colon X_n \to X_{n+1}$, which satisfy the simplicial identities.

The standard n-simplex is the representable functor

$$\Delta[n]_m = \mathrm{Hom}_{\Delta}([m],[n]),$$

so by Yoneda a map $\Delta[n] \to X$ is the same thing as an n-simplex of X. Inside it sit the boundary $\partial\Delta[n]$, the union of all proper faces, and for $0 \le k \le n$ the horn $\Lambda^k[n]$, the union of all codimension-one faces except the kth:

$$\Lambda^k[n] \subseteq \partial\Delta[n] \subseteq \Delta[n].$$

A map $\Lambda^k[n] \to X$ is an n-simplex of X with its interior and one face missing. A Kan complex is a simplicial set in which every horn can be filled.

5.2 Categories enriched in simplicial sets

Definition 5.2 (Simplicially enriched category)**.** A category $\mathcal{C}$ enriched in simplicial sets has objects and, for each pair X, Y, a simplicial set $\mathrm{Map}_{\mathcal{C}}(X,Y)$, with composition and unit maps

$$\mathrm{Map}_{\mathcal{C}}(Y,Z) \times \mathrm{Map}_{\mathcal{C}}(X,Y) \longrightarrow \mathrm{Map}_{\mathcal{C}}(X,Z), \qquad \Delta[0] \longrightarrow \mathrm{Map}_{\mathcal{C}}(X,X),$$

that are associative and unital as maps of simplicial sets.

The vertices of the mapping spaces are the morphisms of an ordinary category,

$$\mathrm{Hom}_{\mathcal{C}}(X,Y) = \mathrm{Map}_{\mathcal{C}}(X,Y)_0.$$

A 1-simplex is a homotopy between two maps, a 2-simplex is a homotopy between homotopies, and so on. For this reading to make sense the mapping spaces should be Kan complexes, which is what the next subsection arranges.

5.3 Tensor, cotensor, and the SM7 axiom

Definition 5.3 (Tensor and cotensor)**.** A simplicially enriched category $\mathcal{M}$ is *tensored* and *cotensored* over $\mathbf{sSet}$ if for $X, Y \in \mathcal{M}$ and $K \in \mathbf{sSet}$ there are objects $X \otimes K$ and Y^K with natural bijections

$$\mathcal{M}(X \otimes K, Y) \cong \mathbf{sSet}(K, \mathrm{Map}_{\mathcal{M}}(X,Y)) \cong \mathcal{M}(X, Y^K). \tag{5.1}$$

For example, a 1-simplex of $\mathrm{Map}_{\mathcal{M}}(X,Y)$ is a map $X \otimes \Delta[1] \to Y$. In a simplicial model category $X \otimes \Delta[1]$ is a cylinder object when X is cofibrant, and $Y^{\Delta[1]}$ is a path object when Y is fibrant, so this matches the earlier notion of homotopy.

Definition 5.4 (Simplicial model category and SM7)**.** A *simplicial model category* is a model category $\mathcal{M}$ that is enriched, tensored, and cotensored over **sSet** and satisfies the following axiom, called *SM7*. For a cofibration $i\colon A \to B$ and a fibration $p\colon X \to Y$, the induced map

$$\operatorname{Map}_{\mathcal{M}}(B, X) \longrightarrow \operatorname{Map}_{\mathcal{M}}(A, X) \underset{\operatorname{Map}_{\mathcal{M}}(A,Y)}{\times} \operatorname{Map}_{\mathcal{M}}(B, Y) \tag{5.2}$$

is a fibration of simplicial sets, and it is acyclic if i or p is.

A vertex of the target of (5.2) is a commutative square

$$\begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle p} \\ B & \longrightarrow & Y, \end{array}$$

and a vertex of the source lying over it is a lift. So SM7 is the lifting axiom (MC3) upgraded from sets of lifts to spaces of lifts. Through (5.1) it is equivalent to the pushout–product form: for a cofibration $i\colon A \to B$ in $\mathcal{M}$ and a cofibration $j\colon K \to L$ in **sSet**, the map

$$(B \otimes K) \amalg_{A \otimes K} (A \otimes L) \longrightarrow B \otimes L \tag{5.3}$$

is a cofibration, acyclic if i or j is.

Proposition 5.5 (Mapping spaces between good objects)**.** *If A is cofibrant and X is fibrant in a simplicial model category, then $\operatorname{Map}_{\mathcal{M}}(A, X)$ is a Kan complex and*

$$\operatorname{Ho}(\mathcal{M})(A, X) \cong \pi_0 \operatorname{Map}_{\mathcal{M}}(A, X).$$

For general A and X the derived mapping space is

$$\operatorname{Map}^h_{\mathcal{M}}(A, X) := \operatorname{Map}_{\mathcal{M}}(QA, RX),$$

well defined up to weak equivalence.

Idea. Apply SM7 to $\varnothing \to A$ and $X \to *$: the target of (5.2) is then a point, so $\operatorname{Map}(A, X) \to *$ is a Kan fibration. The statement about π_0 is Proposition 2.2. □

5.4 Simplicial localization

Dwyer and Kan showed that mapping spaces can be built from the weak equivalences alone, with no model structure.

Construction 5.6 (Hammock localization)**.** For a category with weak equivalences $(\mathcal{C}, W)$, the hammock localization $L^H_W\mathcal{C}$ is a simplicially enriched category with the same objects as $\mathcal{C}$. A k-simplex of the mapping space from X to Y is a "hammock": $k+1$ zigzags from X to Y of the same shape, stacked on top of each other and connected by vertical maps, where the backward arrows and the vertical arrows are in W. Composition concatenates hammocks.

On π_0 this recovers the ordinary localization,

$$\pi_0 \operatorname{Map}_{L^H_W\mathcal{C}}(X, Y) \cong \mathcal{C}[W^{-1}](X, Y),$$

and for a simplicial model category it recovers the derived mapping spaces,

$$\operatorname{Map}_{L^H_W\mathcal{M}}(X, Y) \simeq \operatorname{Map}_{\mathcal{M}}(QX, RY).$$

In particular the derived mapping spaces depend only on $(\mathcal{M}, W)$.

6 $(\infty, 1)$-Categories and Quasicategories

6.1 Models of $(\infty, 1)$-categories

Definition 6.1 ($(\infty, 1)$-category)**.** Informally, an $(\infty, 1)$-*category* has objects, 1-morphisms, 2-morphisms between those, and so on, where all k-morphisms with $k > 1$ are invertible up to higher morphisms. The 1-morphisms need not be invertible.

There are several ways to make this precise: simplicial categories, complete Segal spaces, and quasicategories are the common ones, and they are all equivalent. We use quasicategories. Two extreme cases help to fix ideas. An ordinary category is an $(\infty, 1)$-category with discrete mapping spaces, and a Kan complex is one in which the 1-morphisms are invertible as well, that is, an ∞-groupoid. Both sit inside quasicategories:

$$\mathbf{Cat} \xhookrightarrow{N} \mathbf{qCat} \xhookleftarrow[\text{all edges invertible}]{} \mathbf{Kan}.$$

6.2 Inner horns and coherent composition

Definition 6.2 (Quasicategory)**.** A *quasicategory* is a simplicial set $\mathcal{C}$ in which every inner horn has a filler: for $n \geq 2$ and $0 < k < n$, every diagram

$$\begin{array}{ccc} \Lambda^k[n] & \longrightarrow & \mathcal{C} \\ \downarrow & \nearrow & \\ \Delta[n] & & \end{array}$$

has a dashed lift. The filler is not required to be unique.

The vertices of $\mathcal{C}$ are its objects and the edges are its morphisms. The simplest inner horn is $\Lambda^1[2]$, which is a pair of composable edges:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow^{h} & \downarrow{\scriptstyle g} \\ & & z. \end{array}$$

A filler is a 2-simplex with boundary (f, g, h); it chooses a composite h and records the homotopy $h \simeq g \circ f$. The composite is not unique, but the fillers of the higher inner horns say that any two choices are homotopic, that these homotopies are themselves unique up to homotopy, and so on. The outer horns $\Lambda^0[n]$ and $\Lambda^n[n]$ ask for something different: filling $\Lambda^0[2]$ amounts to dividing by f, so requiring outer fillers as well forces every edge to be invertible.

A diagram of shape K in $\mathcal{C}$ is a map of simplicial sets $K \to \mathcal{C}$, and diagrams form a simplicial set

$$\mathrm{Fun}(K, \mathcal{C}) = \mathcal{C}^K, \qquad (\mathcal{C}^K)_n = \mathbf{sSet}(K \times \Delta[n], \mathcal{C}),$$

which is again a quasicategory.

Proposition 6.3. *Every Kan complex is a quasicategory, and a quasicategory is a Kan complex exactly when all its edges are equivalences. The nerve $N\mathcal{A}$ of an ordinary category is a quasicategory with unique inner-horn fillers, and it is a Kan complex exactly when $\mathcal{A}$ is a groupoid.*

6.3 Mapping spaces, the core, and the homotopy category

A quasicategory has mapping spaces $\mathrm{Map}_{\mathcal{C}}(x, y)$. There are several constructions, all giving equivalent Kan complexes, and I will not distinguish between them.

Definition 6.4 (Homotopy category of a quasicategory)**.** The homotopy category $h\mathcal{C}$ of a quasicategory $\mathcal{C}$ has the vertices of $\mathcal{C}$ as objects and

$$\mathrm{Hom}_{h\mathcal{C}}(x, y) = \pi_0 \,\mathrm{Map}_{\mathcal{C}}(x, y)$$

as morphisms. Composition is defined by filling inner 2-horns, and the 3-dimensional fillers show that it is well defined and associative.

An edge of $\mathcal{C}$ is an *equivalence* if it becomes an isomorphism in $h\mathcal{C}$.

Definition 6.5 (Core)**.** The *core* $\mathcal{C}^{\simeq}$ is the largest Kan complex inside $\mathcal{C}$: it has all the objects, and exactly those edges that are equivalences.

It is the ∞-groupoid of objects of $\mathcal{C}$ and equivalences between them. For the nerve of a category it is the nerve of the maximal subgroupoid, and for a Kan complex it is everything.

6.4 The Joyal model structure

Theorem 6.6 (Joyal model structure)**.** *There is a model structure on* **sSet** *whose cofibrations are the monomorphisms and whose fibrant objects are the quasicategories. Its weak equivalences are the categorical equivalences; between quasicategories these are the maps that are essentially surjective on homotopy categories and induce weak equivalences on all mapping spaces.*

So **sSet** carries two model structures with the same cofibrations. The Kan–Quillen one models spaces, and the Joyal one models $(\infty, 1)$-categories.

6.5 The underlying ∞-category of a model category

A model category $(\mathcal{M}, W)$ has an underlying ∞-category $\mathcal{M}_\infty$. In general one takes the simplicial localization and converts it to a quasicategory. If $\mathcal{M}$ is a simplicial model category there is a shortcut: the simplicial category $\mathcal{M}^{\mathrm{cf}}$ of bifibrant objects has Kan mapping spaces by Proposition 5.5, and its homotopy-coherent nerve is a quasicategory.

Proposition 6.7 (Comparison with the model category)**.** *For a simplicial model category $\mathcal{M}$ and $X, Y \in \mathcal{M}$,*

$$\mathrm{Map}_{\mathcal{M}_\infty}(X, Y) \simeq \mathrm{Map}_{\mathcal{M}}(QX, RY),$$

and

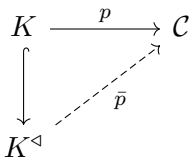
$$h(\mathcal{M}_\infty) \simeq \mathrm{Ho}(\mathcal{M}).$$

6.6 Joins, cones, and ∞-categorical limits

To talk about limits we need cones. The join with a point adds a cone vertex to a simplicial set K, on the left or on the right:

$$K^{\triangleleft} = \Delta[0] \star K, \qquad K^{\triangleright} = K \star \Delta[0].$$

Given a diagram $p\colon K \to \mathcal{C}$, a cone on p with vertex x is an extension



that sends the new vertex to x; a cocone uses $K^{\triangleright}$ instead. The cones on p form a quasicategory, namely the fiber

$$\mathrm{Cone}(p) = \mathrm{Fun}(K^{\triangleleft}, \mathcal{C}) \underset{\mathrm{Fun}(K,\mathcal{C})}{\times} \{p\},$$

and similarly for cocones.

Definition 6.8 (∞-categorical limit and colimit)**.** A *limit* of p is a terminal object of $\mathrm{Cone}(p)$; its cone vertex is written $\lim_K p$. A *colimit* is an initial object of the quasicategory of cocones, with vertex $\mathrm{colim}_K\, p$.

In terms of mapping spaces, ℓ is a limit and c is a colimit of p when there are natural equivalences

$$\mathrm{Map}_{\mathcal{C}}(x, \ell) \simeq \mathrm{Map}_{\mathrm{Fun}(K,\mathcal{C})}(\Delta x, p), \tag{6.1}$$

$$\mathrm{Map}_{\mathcal{C}}(c, y) \simeq \mathrm{Map}_{\mathrm{Fun}(K,\mathcal{C})}(p, \Delta y), \tag{6.2}$$

where Δx and Δy are constant diagrams. This is the usual universal property with hom-sets replaced by mapping spaces, and a limit in this sense is generally not a limit in the homotopy category $h\mathcal{C}$.

For the span and cospan shapes these are the homotopy-coherent pushout and pullback. More generally, the limits and colimits in $\mathcal{M}_\infty$ are the homotopy limits and colimits of $\mathcal{M}$ from (4.2):

$$\operatorname*{colim}_{\mathcal{M}_\infty} D \simeq \operatorname*{hocolim}_{\mathcal{M}} D, \qquad \lim_{\mathcal{M}_\infty} D \simeq \operatorname*{holim}_{\mathcal{M}} D.$$

Part II

Day 2: Spectra and Stable Categories

Day 2 was about spectra. The first half went through point-set models (sequential, symmetric, orthogonal, Γ-spaces) and what each of them does for the smash product. The second half was the abstract side: stable model categories, stabilization of an ∞-category, and the triangulated and t-structure language on the homotopy category. As before I mostly suppress cofibrancy conditions; [MMSS01] is the reference for diagram spectra and [Lur17] for the ∞-categorical statements.

7 Sequential Spectra and Smash Products

7.1 Sequential spectra

Everything starts from the suspension–loop adjunction on pointed spaces,

$$\Sigma = - \wedge S^1 \colon \mathbf{Top}_* \rightleftarrows \mathbf{Top}_* \colon \Omega = \mathrm{Map}_*(S^1, -). \tag{7.1}$$

Definition 7.1 (Sequential spectrum)**.** A *sequential spectrum* X is a sequence of pointed spaces X_n, $n \geq 0$, with structure maps

$$\sigma_n \colon S^1 \wedge X_n \longrightarrow X_{n+1}.$$

A morphism $X \to Y$ is a sequence of pointed maps $X_n \to Y_n$ commuting with the structure maps. By (7.1) the structure maps can equally be given as

$$\widetilde{\sigma}_n \colon X_n \longrightarrow \Omega X_{n+1}.$$

The basic example is the suspension spectrum of a pointed space K,

$$(\Sigma^\infty K)_n = S^n \wedge K,$$

and in particular the *sphere spectrum*

$$\mathbb{S} = \Sigma^\infty S^0 = (S^0, S^1, S^2, \ldots).$$

A spectrum is an Ω-*spectrum* if all the adjoint structure maps $X_n \to \Omega X_{n+1}$ are weak equivalences.

The stable homotopy groups of X are

$$\pi_k X = \operatorname*{colim}_n \pi_{k+n}(X_n), \tag{7.2}$$

where the colimit is along the structure maps. Here k can be negative. A stable equivalence is a map inducing isomorphisms on all π_k. For an Ω-spectrum the colimit is constant, so $\pi_k X = \pi_{k+n} X_n$ for any n with $k + n \geq 0$.

Spectra represent cohomology theories. An Ω-spectrum E defines a reduced cohomology theory by

$$E^q(K) = [\Sigma^\infty K, \Sigma^q E],$$

which for $q \geq 0$ is just $[K, E_q]_*$. In the other direction, Brown representability gives a representing space for each degree of a reduced cohomology theory, and the suspension isomorphisms make these spaces into an Ω-spectrum.

7.2 Pointed smash products and reduced suspension

For pointed spaces (X, x_0) and (Y, y_0),

$$X \wedge Y = \frac{X \times Y}{X \times \{y_0\} \cup \{x_0\} \times Y}. \tag{7.3}$$

With $Y = S^1 = I/\partial I$ this is the reduced suspension

$$\Sigma X = X \wedge S^1 \cong \frac{X \times I}{X \times \{0,1\} \cup \{x_0\} \times I}. \tag{7.4}$$

Since $S^m \wedge S^n \cong S^{m+n}$, iterating the structure maps of a spectrum gives maps $S^p \wedge X_n \to X_{n+p}$.

7.3 The smash-product problem

We want ring spectra, so we want a smash product of spectra for which a multiplication

$$\mu \colon R \wedge R \longrightarrow R$$

and a unit $\mathbb{S} \to R$ make sense, and we want it to be associative and commutative before passing to the homotopy category.

For sequential spectra this fails. The obvious attempt,

$$(X \wedge_{\mathrm{lev}} Y)_n = X_n \wedge Y_n,$$

has the wrong grading, since $X_n \wedge Y_n$ should live in level $2n$. One can fix the grading by using all $X_p \wedge Y_q$ with $p + q = n$, but then the two ways of moving a suspension coordinate past X_p differ by a permutation of sphere coordinates, and a sequential spectrum has no way to keep track of that. The fix in diagram spectra is to build these permutations into the indexing category.

8 Diagram Spectra

8.1 Enriched indexing categories

Let $\mathcal{V}$ be pointed spaces or pointed simplicial sets. A diagram spectrum is an enriched functor

$$X \colon \mathcal{D} \longrightarrow \mathcal{V}$$

from a small $\mathcal{V}$-enriched category $\mathcal{D}$; different choices of $\mathcal{D}$ give the different kinds of spectra.

Sequential spectra come from the category $\mathcal{H}$ whose objects are the nonnegative integers and whose mapping spaces are

$$\mathcal{H}(m,n) = \begin{cases} S^{n-m}, & m \le n, \\ *, & m > n, \end{cases} \tag{8.1}$$

with composition $S^{n-m} \wedge S^{m-\ell} \cong S^{n-\ell}$. Here $\mathcal{H}(m,n)$ is a pointed space, because $\mathcal{H}$ is enriched over pointed spaces. An enriched functor $X\colon \mathcal{H} \to \mathbf{Top}_*$ consists of spaces X_n and maps

$$\mathcal{H}(m,n) \wedge X_m \longrightarrow X_n,$$

which for $n = m+1$ are the structure maps and for larger gaps are their iterates.

If $\mathcal{D}$ is symmetric monoidal, the functor category gets a symmetric monoidal product by Day convolution:

$$(X \star Y)(d) = \int^{a,b\in\mathcal{D}} \mathcal{D}(a \otimes b, d) \otimes X(a) \otimes Y(b). \tag{8.2}$$

Its unit is the representable functor $\mathcal{D}(\mathbb{1}, -)$, and it has an internal hom:

$$[\mathcal{D},\mathcal{V}](X \star Y, Z) \cong [\mathcal{D},\mathcal{V}](X, \underline{\mathrm{Hom}}(Y,Z)).$$

The category $\mathcal{H}$ is monoidal under addition but not symmetric, which is the smash-product problem again.

8.2 Coends, tensors, and Kan extension

The integral sign in (8.2) is a coend. For a functor $F\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{V}$ it is the coequalizer

$$\int^{c\in\mathcal{C}} F(c,c) = \mathrm{coeq}\left(\coprod_{u\colon c\to c'} F(c',c) \rightrightarrows \coprod_{c} F(c,c) \right), \tag{8.3}$$

where the two maps let u act on the first and on the second variable.

The example to have in mind is the tensor product of modules. A ring R is a one-object category enriched in abelian groups, a right module M and a left module N are functors of opposite variance, and

$$M \otimes_R N \cong \int^{*} M(*) \otimes_{\mathbb{Z}} N(*).$$

The coequalizer imposes $(mr, n) \sim (m, rn)$. So a coend is a tensor product over a category with many objects.

Left Kan extension is given by the same formula. For an enriched functor $i\colon \mathcal{D} \to \mathcal{E}$,

$$(\mathrm{Lan}_i X)(e) = \int^{d\in\mathcal{D}} \mathcal{E}(id, e) \otimes X(d), \tag{8.4}$$

and Lan_i is left adjoint to restriction i^*. When X is representable the coend collapses by the Yoneda lemma, so $\mathrm{Lan}_i\, \mathcal{D}(d,-) = \mathcal{E}(id,-)$.

8.3 Symmetric spectra

Definition 8.1 (Symmetric spectrum)**.** A *symmetric spectrum* X consists of pointed Σ_n-spaces X_n for $n \geq 0$ and structure maps

$$\sigma_n \colon X_n \wedge S^1 \longrightarrow X_{n+1}$$

such that each iterate

$$X_n \wedge S^p \longrightarrow X_{n+p} \tag{8.5}$$

is $\Sigma_n \times \Sigma_p$-equivariant, where $\Sigma_n \times \Sigma_p \subseteq \Sigma_{n+p}$ acts by block permutations.

The Σ_n-action on X_n is there to match the permutation action on the coordinates of $S^n = (S^1)^{\wedge n}$.

The smash product is built in two steps. A *symmetric sequence* is just the spaces X_n with their Σ_n-actions, without structure maps. Symmetric sequences have a tensor product

$$(A \otimes B)_n = \bigvee_{p+q=n} (\Sigma_n)_+ \wedge_{\Sigma_p \times \Sigma_q} (A_p \wedge B_q), \tag{8.6}$$

where for a subgroup $H \subseteq G$ and an H-space Z the induced G-space

$$\operatorname{Ind}_H^G Z = G_+ \wedge_H Z$$

consists of pairs (g, z) modulo $(gh, z) \sim (g, hz)$. This is the "sum over $p + q = n$" from the previous section, with the induction supplying the missing permutations. The sphere sequence $\mathbb{S}_n = S^n$ is a commutative monoid for $\otimes$, and a symmetric spectrum is the same thing as a module over it. The smash product of symmetric spectra is then the tensor product of modules over $\mathbb{S}$:

$$X \wedge_{\mathbb{S}} Y = \operatorname{coeq}\big(X \otimes \mathbb{S} \otimes Y \rightrightarrows X \otimes Y\big), \tag{8.7}$$

with unit $\mathbb{S}$.

Evaluation at level n, viewed as a functor to Σ_n-spaces, has a left adjoint F_n. For a Σ_n-space K the free spectrum $F_n K$ is trivial below level n and

$$(F_n K)_{n+p} \cong (\Sigma_{n+p})_+ \wedge_{\Sigma_n \times \Sigma_p} (K \wedge S^p) \tag{8.8}$$

above it.

On the homotopy-theoretic side, the fibrant objects of the stable model structure are the levelwise fibrant Ω-spectra, and the smash product is compatible with the model structure in the sense of the pushout–product axiom: for cofibrations $i \colon A \to B$ and $j \colon K \to L$,

$$i \,\square\, j \colon (B \wedge K) \amalg_{A \wedge K} (A \wedge L) \longrightarrow B \wedge L \tag{8.9}$$

is a cofibration, and it is acyclic if i or j is. So the derived smash product is computed by $QX \wedge QY$.

The model structure can be transferred to associative monoids and to their modules. Commutative monoids are trickier, because of symmetric powers of objects in level zero. The fix is the *positive* stable model structure, which comes back on Day 3.

8.4 Orthogonal spectra and Thom mapping spaces

Orthogonal spectra replace the index n by a finite-dimensional real inner-product space V, and Σ_n by the orthogonal group $O(V)$. Let $\mathcal{L}(V,W)$ be the space of linear isometric embeddings $V \hookrightarrow W$. Over it there is the complement bundle

$$\xi(V,W) = \{(f,w) \mid f \in \mathcal{L}(V,W),\ w \perp f(V)\},$$

whose fiber at f is the orthogonal complement $W - f(V)$. The mapping spaces of the indexing category $\mathcal{O}$ are the Thom spaces

$$\mathcal{O}(V,W) = \mathrm{Th}(\xi(V,W)), \tag{8.10}$$

where for a vector bundle $E \to B$

$$\mathrm{Th}(E) = D(E)/S(E)$$

is the disk bundle modulo the sphere bundle, or equivalently the fiberwise one-point compactification with all the points at infinity identified. Composition in $\mathcal{O}$ composes the isometries and adds the vectors in the complements, and direct sum makes $\mathcal{O}$ symmetric monoidal. As a check, $\mathcal{O}(0,W) \cong S^W$.

Definition 8.2 (Orthogonal spectrum)**.** An *orthogonal spectrum* is a continuous functor $X\colon \mathcal{O} \to \mathbf{Top}_*$. Unwinding, it has based $O(V)$-spaces $X(V)$ and compatible structure maps

$$X(V) \wedge S^W \longrightarrow X(V \oplus W).$$

The smash product is Day convolution over $\mathcal{O}$. Orthogonal spectra, symmetric spectra, and the S-modules of EKMM are related by monoidal Quillen equivalences, so they give the same theory of ring spectra.

8.5 The excisive-functor model

A fourth description uses no indexing at all: a spectrum is a reduced excisive functor from finite pointed CW complexes to spaces. A spectrum E gives the functor

$$K \longmapsto \Omega^\infty(E \wedge \Sigma^\infty K). \tag{8.11}$$

To get from such a functor back to a spectrum, recall that *excisive* means that homotopy pushout squares are sent to homotopy pullback squares. Applying this to the suspension square

$$\begin{array}{ccc} K & \longrightarrow & * \\ \downarrow & & \downarrow \\ * & \longrightarrow & \Sigma K \end{array}$$

gives $F(K) \simeq \Omega F(\Sigma K)$ for a reduced excisive F, so the spaces $F(S^n)$ form an Ω-spectrum.

9 Γ-Spaces and Connective Spectra

9.1 Special and very special Γ-spaces

Let Γ be the category of finite pointed sets

$$n_+ = \{0, 1, \dots, n\},$$

with basepoint 0. In these notes a Γ-space is a pointed functor $A\colon \Gamma \to \mathbf{Top}_*$. (Some authors call the opposite category Γ and then write Γ-spaces contravariantly.) For $1 \le i \le n$ let $p_i\colon n_+ \to 1_+$ send i to 1 and everything else to 0. Together these induce the *Segal map*

$$A(n_+) \longrightarrow A(1_+)^n. \tag{9.1}$$

Definition 9.1 (Γ-space)**.** A Γ-space A is *special* if $A(0_+) \simeq *$ and every Segal map (9.1) is a weak equivalence. It is *very special* if moreover the commutative monoid $\pi_0 A(1_+)$, with the operation induced by the fold map $2_+ \to 1_+$, is a group.

So a special Γ-space is a space $A(1_+)$ with a multiplication that is commutative and associative up to all higher homotopies, and "very special" says that it has inverses up to homotopy.

A Γ-space gives a spectrum. Extend A from finite pointed sets to pointed spaces and evaluate on spheres:

$$\mathbf{B}A = (A(S^0), A(S^1), A(S^2), \dots), \tag{9.2}$$

with structure maps given by the assembly maps $S^1 \wedge A(S^n) \to A(S^{n+1})$. The spectra obtained this way are connective, and very special Γ-spaces model all connective spectra.

9.2 Smash product and model structures

Finite pointed sets are symmetric monoidal under the smash product, with unit 1_+, so Day convolution gives a smash product of Γ-spaces:

$$(A \wedge_\Gamma B)(n_+) = \int^{k_+, \ell_+ \in \Gamma} \underline{\Gamma}(k_+ \wedge \ell_+, n_+) \wedge A(k_+) \wedge B(\ell_+). \tag{9.3}$$

The hom-sets are based at the constant map, and the unit is $\underline{\Gamma}(1_+, -)$.

The model structures are obtained by localizing the objectwise one in two steps, first at the Segal maps and then at the group-completion map. The three stages are

$$\text{diagrams} \to \text{homotopy-coherent monoids} \to \text{grouplike monoids} \simeq \text{connective spectra}.$$

9.3 Algebraic K-theory

This is where K-theory spectra come from. A permutative category $\mathcal{C}$ gives a special Γ-space, and the associated spectrum is $K(\mathcal{C})$, with

$$\Omega^\infty K(\mathcal{C}) \simeq (B\mathcal{C})^{\mathrm{gp}},$$

the group completion of the classifying space. Some examples:

1. Finite sets and bijections under disjoint union give the sphere spectrum. This is the Barratt–Priddy–Quillen theorem.
2. Finitely generated projective R-modules and isomorphisms under direct sum give the algebraic K-theory spectrum $K(R)$.
3. Finite-dimensional complex (or real) vector spaces and linear isomorphisms, as a topological groupoid, give connective topological K-theory ku (or ko).

10 Stable Model Categories and Spectral Enrichment

10.1 Stability as a Quillen condition

Let $\mathcal{M}$ be a pointed model category. If $\mathcal{M}$ is simplicial or topological, tensoring and cotensoring with S^1 give a Quillen adjunction

$$\Sigma\colon \mathcal{M} \rightleftarrows \mathcal{M}\colon \Omega.$$

Without an enrichment one defines ΣX as the homotopy cofiber of $X \to *$ and ΩX as the homotopy fiber of $* \to X$, and these still give an adjunction on $\mathrm{Ho}(\mathcal{M})$.

Definition 10.1 (Stable model category)**.** A pointed model category $\mathcal{M}$ is *stable* if

$$\mathbf{L}\Sigma\colon \mathrm{Ho}(\mathcal{M}) \rightleftarrows \mathrm{Ho}(\mathcal{M})\colon \mathbf{R}\Omega$$

is an equivalence.

Spectra, unbounded chain complexes, and modules over a ring spectrum are stable. In a stable model category a square is a homotopy pushout if and only if it is a homotopy pullback.

10.2 SM7 and enrichment

Recall from Day 1 that SM7 says that the map (5.2) of mapping spaces is a Kan fibration, and that as a result $\mathrm{Map}(A, X)$ is a Kan complex computing the derived mapping space when A is cofibrant and X is fibrant. In a stable setting we would like mapping *spectra*, and the definition is the same with simplicial sets replaced by spectra.

Definition 10.2 (Spectral model category)**.** A *spectral model category* is a model category $\mathcal{M}$ that is enriched, tensored, and cotensored over a symmetric monoidal model category of spectra,

$$\mathcal{M}(K \wedge X, Y) \cong \mathbf{Sp}(K, \underline{\mathrm{Map}}_{\mathcal{M}}(X, Y)) \cong \mathcal{M}(X, Y^K),$$

and satisfies the analogue of SM7.

For cofibrant X and fibrant Y the mapping spectrum knows the maps of all degrees:

$$\pi_n \underline{\mathrm{Map}}_{\mathcal{M}}(X, Y) \cong \mathrm{Ho}(\mathcal{M})(\Sigma^n X, Y), \qquad n \in \mathbb{Z}. \tag{10.1}$$

10.3 Spectral categories

Definition 10.3 (Spectral category)**.** A *spectral category* $\mathcal{A}$ is a category enriched in spectra: for objects a, b there is a mapping spectrum $\mathcal{A}(a, b)$, with unit maps $\mathbb{S} \to \mathcal{A}(a, a)$ and associative composition maps

$$\mathcal{A}(b, c) \wedge \mathcal{A}(a, b) \longrightarrow \mathcal{A}(a, c).$$

A spectral category with one object is a ring spectrum, so spectral categories are "ring spectra with many objects." Accordingly, a module over $\mathcal{A}$ is an enriched functor $\mathcal{A}^{\mathrm{op}} \to \mathbf{Sp}$, and $\pi_0\mathcal{A}(a, b)$ is the set of morphisms in the homotopy category of $\mathcal{A}$.

The reason to care is Morita theory on Day 3, where spectral categories show up as endomorphism categories of sets of generators.

10.4 Spectra internal to a model category

One can also form spectra in a model category other than spaces. If $\mathcal{M}$ is pointed and tensored over pointed simplicial sets, a symmetric spectrum in $\mathcal{M}$ is a sequence of objects $X_n \in \mathcal{M}$ with Σ_n-actions and equivariant maps

$$S^1 \otimes X_n \longrightarrow X_{n+1}.$$

The stable model structure $\mathbf{Sp}^\Sigma(\mathcal{M})$ is the localization of the levelwise one that makes Ω-spectra the fibrant objects. Evaluation at level zero has a left adjoint,

$$\Sigma^\infty_{\mathcal{M}} \colon \mathcal{M} \rightleftarrows \mathbf{Sp}^\Sigma(\mathcal{M}) \colon \mathrm{Ev}_0, \tag{10.2}$$

and this adjunction is a Quillen equivalence when $\mathcal{M}$ is already stable. So stabilizing twice does nothing new.

11 Stabilization in ∞-Categories

11.1 Excisive functors as spectrum objects

In an ∞-category the excisive-functor description becomes the definition. Let $\mathcal{C}$ be an ∞-category with finite limits. A functor

$$F \colon \mathcal{S}_*^{\mathrm{fin}} \longrightarrow \mathcal{C}$$

from finite pointed spaces is *reduced* if $F(*)$ is terminal and *excisive* if it sends pushout squares to pullback squares.

Definition 11.1 (Stabilization)**.** The ∞-category of spectrum objects in $\mathcal{C}$ is

$$\mathbf{Sp}(\mathcal{C}) = \mathrm{Exc}_*(\mathcal{S}_*^{\mathrm{fin}}, \mathcal{C}), \tag{11.1}$$

the full subcategory of reduced excisive functors. For the ∞-category $\mathcal{S}$ of spaces we write $\mathbf{Sp} = \mathbf{Sp}(\mathcal{S})$.

As before, excision gives $F(S^n) \simeq \Omega F(S^{n+1})$, so a spectrum object amounts to a sequence

$$(X_0, X_1, \ldots), \qquad X_n \xrightarrow{\simeq} \Omega X_{n+1}. \tag{11.2}$$

When $\mathcal{C}$ is presentable there is an adjunction

$$\Sigma^\infty \colon \mathcal{C} \rightleftarrows \mathbf{Sp}(\mathcal{C}) \colon \Omega^\infty, \tag{11.3}$$

in which Ω^∞ is evaluation at S^0. If $\mathcal{C}$ is not pointed, one first adds a disjoint basepoint and writes Σ^∞_+.

11.2 Symmetric monoidal stabilization and Day convolution

The smash product on $\mathbf{Sp}$ is again a Day convolution, now of excisive functors, and its unit is $\mathbb{S}$. More generally, if $\mathcal{C}$ is presentably symmetric monoidal (the tensor product preserves colimits in each variable), then $\mathbf{Sp}(\mathcal{C})$ is symmetric monoidal as well, and we can form

$$\mathrm{CAlg}(\mathbf{Sp}(\mathcal{C})), \qquad \mathrm{Alg}(\mathbf{Sp}(\mathcal{C})), \qquad \mathrm{Mod}_R(\mathbf{Sp}(\mathcal{C})).$$

Homotopy groups are maps out of shifts of the unit:

$$\pi_n X \cong \mathrm{Ho}(\mathbf{Sp})(\Sigma^n \mathbb{S}, X).$$

Understanding / Intuition

The addition on $\pi_* R$ comes for free: **Sp** *is stable, so every spectrum has it. What the map $R \wedge R \to R$ of a ring spectrum adds is the multiplication, and with it the products $\pi_p R \otimes \pi_q R \to \pi_{p+q} R$.*

Stabilization has a universal property. For $\mathcal{C}$ pointed and presentable and $\mathcal{D}$ stable and presentable, composing with Σ^∞ gives an equivalence

$$\mathrm{Fun}^L(\mathbf{Sp}(\mathcal{C}), \mathcal{D}) \simeq \mathrm{Fun}^L(\mathcal{C}, \mathcal{D})$$

between categories of colimit-preserving functors. So $\mathbf{Sp}(\mathcal{C})$ is the universal stable category receiving a map from $\mathcal{C}$; see [Lur17, Sections 1.4 and 4.8]. If $\mathcal{C}$ is already stable then $\Omega^\infty \colon \mathbf{Sp}(\mathcal{C}) \to \mathcal{C}$ is an equivalence.

11.3 Stable ∞-categories

Definition 11.2 (Stable ∞-category)**.** An ∞-category $\mathcal{C}$ is *stable* if

1. it has a zero object;
2. it has finite limits and finite colimits; and
3. a square in $\mathcal{C}$ is a pushout if and only if it is a pullback.

The cofiber and fiber of a map $f \colon X \to Y$ are defined by the squares

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{cof}(f) \end{array} \qquad \begin{array}{ccc} \mathrm{fib}(f) & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle f} \\ 0 & \longrightarrow & Y, \end{array}$$

and condition (3) gives

$$\mathrm{cof}(f) \simeq \Sigma \mathrm{fib}(f), \qquad \mathrm{fib}(f) \simeq \Omega \mathrm{cof}(f).$$

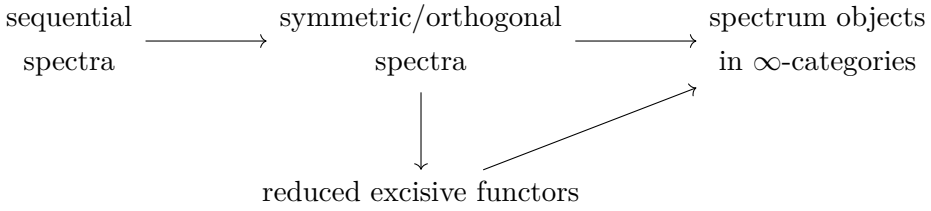
In particular Σ and Ω are inverse equivalences. Finite products and coproducts agree, so there are biproducts $X \oplus Y$.

A stable ∞-category is automatically enriched in spectra. The mapping spaces are infinite loop spaces, and there are mapping spectra $\mathrm{map}_{\mathcal{C}}(X, Y)$ with

$$\Omega^\infty \mathrm{map}_{\mathcal{C}}(X, Y) \simeq \mathrm{Map}_{\mathcal{C}}(X, Y), \qquad \pi_n \mathrm{map}_{\mathcal{C}}(X, Y) \cong \mathrm{Ho}(\mathcal{C})(\Sigma^n X, Y). \tag{11.4}$$

The right notion of functor between stable ∞-categories is an *exact* one, meaning that it preserves finite limits, or equivalently finite colimits.

Here is how the models of spectra fit together:



Each arrow is an equivalence of homotopy theories. On the point-set level the models are quite different, and only the middle ones have a good smash product.

12 Triangulated Categories and t-Structures

12.1 The triangulated homotopy category

Let $\mathcal{C}$ be a stable ∞-category. Its homotopy category $\mathrm{Ho}(\mathcal{C})$ is additive, suspension is an autoequivalence [1] of it, and every morphism $f\colon X \to Y$ sits in a triangle

$$X \xrightarrow{f} Y \longrightarrow \mathrm{cof}(f) \longrightarrow \Sigma X. \tag{12.1}$$

Declaring the triangles isomorphic to these to be distinguished makes $\mathrm{Ho}(\mathcal{C})$ a triangulated category. The axioms, including the octahedral one, are consequences of properties of pushout squares in $\mathcal{C}$.

Mapping (12.1) into an object Z gives a fiber sequence of mapping spectra, and so a long exact sequence

$$\cdots \to \mathrm{Ho}(\mathcal{C})(\Sigma X, Z) \to \mathrm{Ho}(\mathcal{C})(\mathrm{cof}(f), Z) \to \mathrm{Ho}(\mathcal{C})(Y, Z) \to \mathrm{Ho}(\mathcal{C})(X, Z) \to \cdots .$$

Mapping out of Z gives the covariant one.

12.2 t-Structures

Definition 12.1 (t-structure)**.** A t-structure on a stable ∞-category $\mathcal{C}$ is a pair of full subcategories $(\mathcal{C}_{\geq 0}, \mathcal{C}_{\leq 0})$ such that, with

$$\mathcal{C}_{\geq n} = \Sigma^n \mathcal{C}_{\geq 0}, \qquad \mathcal{C}_{\leq n} = \Sigma^n \mathcal{C}_{\leq 0},$$

1. $\mathcal{C}_{\geq 1} \subseteq \mathcal{C}_{\geq 0}$ and $\mathcal{C}_{\leq 0} \subseteq \mathcal{C}_{\leq 1}$;
2. $\mathrm{Map}_{\mathcal{C}}(X, Y)$ is contractible for $X \in \mathcal{C}_{\geq 0}$ and $Y \in \mathcal{C}_{\leq -1}$;
3. every $X \in \mathcal{C}$ fits into a cofiber sequence

$$\tau_{\geq 0} X \longrightarrow X \longrightarrow \tau_{\leq -1} X \tag{12.2}$$

with $\tau_{\geq 0} X \in \mathcal{C}_{\geq 0}$ and $\tau_{\leq -1} X \in \mathcal{C}_{\leq -1}$.

The sequence (12.2) is functorial: $\tau_{\geq 0}$ is right adjoint to the inclusion of $\mathcal{C}_{\geq 0}$, and $\tau_{\leq -1}$ is left adjoint to the inclusion of $\mathcal{C}_{\leq -1}$. Shifting gives $\tau_{\geq n}$ and $\tau_{\leq n}$ for all n. The *heart*

$$\mathcal{C}^{\heartsuit} = \mathcal{C}_{\geq 0} \cap \mathcal{C}_{\leq 0}$$

is always an abelian category.

On spectra the standard t-structure is defined by homotopy groups,

$$\mathbf{Sp}_{\geq 0} = \{X \mid \pi_n X = 0 \text{ for } n < 0\}, \qquad \mathbf{Sp}_{\leq 0} = \{X \mid \pi_n X = 0 \text{ for } n > 0\}.$$

Its heart is the category of abelian groups, with A corresponding to the Eilenberg–Mac Lane spectrum HA. The truncations assemble into the Postnikov tower

$$\cdots \longrightarrow \tau_{\leq n} X \longrightarrow \tau_{\leq n-1} X \longrightarrow \cdots ,$$

whose nth layer is $\Sigma^n H\pi_n X$.

Example 12.2 (Chain complexes)**.** On the derived category of a ring R the standard t-structure is defined in the same way by homology groups, and its heart is the category of R-modules.

A t-structure is extra structure on a stable category, and the same category can carry several.

Part III

Day 3: Rings, Modules, and Coherent Multiplication

Day 3 was algebra in the symmetric monoidal category of spectra. It started with ring spectra, their modules, and Morita theory, and then turned to bar constructions and to operads, which are the way to say "associative (or commutative) up to coherent homotopy." The last part compares all of this with ordinary homological algebra through Eilenberg–Mac Lane spectra.

13 Ring Spectra and Their Algebra

13.1 Monoid objects in spectra

Fix one of the symmetric monoidal point-set models $(\mathbf{Sp}, \wedge, \mathbb{S})$ from Day 2. The guiding analogy is

$$(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z}) \qquad \rightsquigarrow \qquad (\mathbf{Sp}, \wedge, \mathbb{S}) :$$

a ring is a monoid in abelian groups, and a ring spectrum is a monoid in spectra.

Definition 13.1 (Ring spectrum)**.** An *associative ring spectrum* is a spectrum R with a multiplication and a unit

$$\mu \colon R \wedge R \longrightarrow R, \qquad \eta \colon \mathbb{S} \longrightarrow R,$$

such that

$$\begin{array}{ccc} (R \wedge R) \wedge R & \xrightarrow{\alpha} & R \wedge (R \wedge R) \\ {\scriptstyle \mu \wedge \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \mathrm{id} \wedge \mu} \\ R \wedge R & \xrightarrow[\mu]{} & R \end{array} \qquad \begin{array}{ccccc} \mathbb{S} \wedge R & \xrightarrow{\eta \wedge \mathrm{id}} & R \wedge R & \xleftarrow{\mathrm{id} \wedge \eta} & R \wedge \mathbb{S} \\ & {\scriptstyle \lambda_R} \searrow & \downarrow{\scriptstyle \mu} & \swarrow {\scriptstyle \rho_R} & \\ & & R & & \end{array}$$

commute, where α, λ, ρ are the coherence isomorphisms of $\wedge$. It is *commutative* if also

$$\begin{array}{ccc} R \wedge R & \xrightarrow{\tau} & R \wedge R \\ & {\scriptstyle \mu} \searrow & \downarrow{\scriptstyle \mu} \\ & & R \end{array}$$

commutes, where τ is the twist.

These diagrams commute on the nose in the point-set category, and as noted on Day 2 it does not matter which model we use. In ∞-categorical language the same objects are the E_1-algebras and the E_∞-algebras in $\mathbf{Sp}$ [Lur17]; the operads behind these names appear in the section on operads below.

Example 13.2. The sphere $\mathbb{S}$ is a commutative ring spectrum. An ordinary ring A gives the Eilenberg–Mac Lane spectrum HA, with

$$\pi_k(HA) \cong \begin{cases} A, & k = 0, \\ 0, & k \neq 0, \end{cases}$$

which is a ring spectrum, commutative if A is. Other examples are KO, KU, and Thom spectra such as MU.

13.2 Modules, algebras, and commutative algebras

Fix an associative ring spectrum R.

Definition 13.3 (R-module spectrum)**.** A *left R-module* is a spectrum M with an action

$$a \colon R \wedge M \longrightarrow M$$

such that

$$\begin{array}{ccc} (R \wedge R) \wedge M & \xrightarrow{\alpha} & R \wedge (R \wedge M) \\ {\scriptstyle \mu \wedge \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \mathrm{id} \wedge a} \\ R \wedge M & \xrightarrow[a]{} & M \end{array}$$

commutes and $a \circ (\eta \wedge \mathrm{id})$ is the unit isomorphism $\mathbb{S} \wedge M \cong M$. Right modules and bimodules are defined in the same way.

When R is commutative, Mod_R is itself symmetric monoidal under the relative smash product $\wedge_R$, with unit R, so the whole story can be repeated inside it.

Definition 13.4 (R-algebra)**.** Let R be commutative. An *R-algebra* is a monoid in $(\mathrm{Mod}_R, \wedge_R, R)$, that is, an R-module A with maps

$$A \wedge_R A \longrightarrow A, \qquad R \longrightarrow A,$$

and a *commutative R-algebra* is a commutative monoid. We write

$$\mathrm{Alg}_R = \mathrm{Alg}(\mathrm{Mod}_R), \qquad \mathrm{CAlg}_R = \mathrm{CAlg}(\mathrm{Mod}_R).$$

With $R = \mathbb{S}$ these are just ring spectra and commutative ring spectra. There are forgetful functors

$$\mathrm{CAlg}_R \xrightarrow{U} \mathrm{Alg}_R \xrightarrow{U} \mathrm{Mod}_R \xrightarrow{U} \mathbf{Sp},$$

and each has a left adjoint. The one we need most is the free commutative algebra,

$$\mathbb{P}_R \colon \mathrm{Mod}_R \rightleftarrows \mathrm{CAlg}_R : U, \qquad \mathbb{P}_R \dashv U,$$

which is given by the symmetric algebra formula

$$\mathbb{P}_R(M) = \bigvee_{n \geq 0} M^{\wedge_R n} / \Sigma_n, \qquad M^{\wedge_R 0} = R.$$

The adjunction says that a map of commutative algebras $\mathbb{P}_R(M) \to A$ is the same as a map of R-modules $M \to U(A)$. The derived version of the formula uses homotopy orbits $M^{\wedge_R n}_{h\Sigma_n}$ in place of the strict orbits, and the difference between the two is the source of the technical trouble in the next subsection.

13.3 Transferred and positive model structures

The model structures on modules and algebras are transferred from spectra along the free–forgetful adjunctions: a map of algebras is a weak equivalence or a fibration if the underlying map of spectra is one, and the cofibrations are determined by lifting.

For modules and associative algebras this works. The condition that makes it work is the *monoid axiom*, which says that cell complexes built from maps of the form $j \wedge X$, with j an acyclic cofibration and X arbitrary, are still weak equivalences.

For commutative algebras it does not work with the usual stable model structure. The problem is the symmetric powers $M^{\wedge n}/\Sigma_n$, which do not preserve weak equivalences between cofibrant objects when the Σ_n-action is not free. Lewis's obstruction shows that something has to give. If the sphere spectrum were cofibrant and commutative monoids had the expected model structure, then a fibrant replacement of $\mathbb{S}$ as a commutative monoid would have as its zeroth space a strictly commutative topological monoid with the homotopy type of QS^0. That is impossible, because such a monoid is a product of Eilenberg–Mac Lane spaces.

The solution is the *positive* stable model structure. It has the same weak equivalences, but a positive cofibration is required to be an isomorphism in level zero. The sphere is then no longer cofibrant, so the obstruction disappears, and commutative ring spectra get a model structure [MMSS01].

13.4 Tensor, cotensor, and enriched mapping spectra

The category Mod_R is enriched, tensored, and cotensored over spectra. For $K \in \mathbf{Sp}$ and $M, N \in \mathrm{Mod}_R$ there are a tensor $K \wedge M$, a cotensor $F(K,N)$, and a mapping spectrum $\underline{\mathrm{Map}}_R(M,N)$, related by

$$\mathrm{Mod}_R(K \wedge M, N) \cong \mathbf{Sp}\big(K, \underline{\mathrm{Map}}_R(M,N)\big),$$
$$\mathrm{Mod}_R(M, F(K,N)) \cong \mathbf{Sp}\big(K, \underline{\mathrm{Map}}_R(M,N)\big).$$

The composition maps

$$\underline{\mathrm{Map}}_R(N,P) \wedge \underline{\mathrm{Map}}_R(M,N) \longrightarrow \underline{\mathrm{Map}}_R(M,P)$$

make Mod_R a spectral category in the sense of Day 2.

For a right module M and a left module N, the derived smash product $M \wedge^{\mathbf{L}}_R N$ is computed by replacing M or N cofibrantly. Its homotopy groups can be computed by the Künneth spectral sequence

$$\mathrm{Tor}^{\pi_* R}_{p,q}(\pi_* M, \pi_* N) \Longrightarrow \pi_{p+q}(M \wedge^{\mathbf{L}}_R N).$$

14 Morita Theory for Ring Spectra

14.1 Classical Morita theory and bimodules

Morita theory starts from the observation that different rings can have the same category of modules.

Definition 14.1 (Morita equivalence)**.** Rings R and S are *Morita equivalent* if there is an additive equivalence

$$R\text{-}\,\mathrm{Mod} \simeq S\text{-}\,\mathrm{Mod}\,.$$

In the derived version one asks for an equivalence of derived categories of modules instead.

Functors between module categories come from bimodules. An (S, R)-bimodule ${}_SP_R$ gives

$$P \otimes_R -\colon R\text{-Mod} \longrightarrow S\text{-Mod},$$

and it is convenient to think of P as an arrow from R to S. Composition of arrows is the tensor product,

$$R \xrightarrow{{}_SP_R} S \xrightarrow{{}_TQ_S} T \qquad \rightsquigarrow \qquad {}_T(Q \otimes_S P)_R,$$

the identity arrow of R is R as a bimodule over itself, and a Morita equivalence is an invertible arrow.

For ring spectra nothing changes except that tensor products become derived smash products. An (S, R)-bimodule spectrum P gives

$$P \wedge^{\mathbf{L}}_R -\colon \mathrm{Ho}(\mathrm{Mod}_R) \longrightarrow \mathrm{Ho}(\mathrm{Mod}_S),$$

and invertible bimodules give derived Morita equivalences.

14.2 The matrix-ring test case

The standard example is

$$R \simeq_{\mathrm{Mor}} M_n(R)$$

for a ring R and $n \geq 1$. Let $P = R^n$ be the module of column vectors, an $(M_n(R), R)$-bimodule, and let

$$P^\vee = \mathrm{Hom}_R(P, R)$$

be the row vectors, an $(R, M_n(R))$-bimodule. Then

$$P \otimes_R P^\vee \cong \mathrm{End}_R(P) \cong M_n(R), \qquad P^\vee \otimes_{M_n(R)} P \cong R$$

(a column times a row is a matrix, a row times a column is a scalar), so P and $P^\vee$ are inverse to each other. For ring spectra the same argument works with R^n replaced by a finite wedge of copies of R.

Invariants that depend only on the module category, such as algebraic K-theory and THH, therefore cannot distinguish R from $M_n(R)$.

14.3 A stable theory with one compact generator

The matrix example is a special case of something much more general. Let $\mathcal{M}$ be a stable model category and G an object. The endomorphisms of G form a ring spectrum

$$R = \mathbf{R}\underline{\mathrm{End}}_{\mathcal{M}}(G) = \mathbf{R}\underline{\mathrm{Map}}_{\mathcal{M}}(G, G),$$

and there is a derived adjunction

$$\mathrm{Mod}_R^{\mathrm{right}} \underset{\mathbf{R}\underline{\mathrm{Map}}(G,-)}{\overset{-\wedge^{\mathbf{L}}_R G}{\rightleftarrows}} \mathcal{M}.$$

To keep the sides straight: R acts on $\mathbf{R}\underline{\mathrm{Map}}(G, X)$ by precomposition, which is a right action, and on G by evaluation, which is a left action.

Theorem 14.2 (Schwede–Shipley, one generator)**.** *Let $\mathcal{M}$ be a simplicial, cofibrantly generated, proper, stable model category with a compact generator G. Then $\mathcal{M}$ is Quillen equivalent, through a zigzag, to right modules over the endomorphism ring spectrum of G:*

$$\mathcal{M} \simeq_{\mathrm{Q}} \mathrm{Mod}^{\mathrm{right}}_{\mathbf{R}\,\mathrm{End}(G)}.$$

Here G is *compact* if

$$[G, \bigvee_i X_i] \cong \bigoplus_i [G, X_i],$$

and it is a *generator* if $[\Sigma^n G, X] = 0$ for all n implies $X \simeq 0$. So a stable homotopy theory with a single compact generator is determined by one ring spectrum; see [SS03]. The homotopy ring $[\Sigma^* G, G]$ would not be enough for this; one needs the endomorphism ring spectrum itself.

14.4 Several generators: ringoids and spectral categories

If there is a set $\mathcal{G} = \{G_i\}_{i \in I}$ of compact generators and no single one, the ring spectrum is replaced by a ring spectrum with many objects.

Definition 14.3 (Spectral endomorphism category)**.** The *spectral endomorphism category* $\mathcal{O}_\mathcal{G}$ has the generators G_i (chosen bifibrant) as objects and mapping spectra

$$\mathcal{O}_\mathcal{G}(G_i, G_j) = \mathbf{R}\underline{\mathrm{Map}}_\mathcal{M}(G_i, G_j),$$

with composition

$$\mathcal{O}(G_j, G_k) \wedge \mathcal{O}(G_i, G_j) \longrightarrow \mathcal{O}(G_i, G_k).$$

A right $\mathcal{O}_\mathcal{G}$-module is a spectral functor

$$\mathcal{O}_\mathcal{G}^{\mathrm{op}} \longrightarrow \mathbf{Sp}.$$

The general Schwede–Shipley theorem says that $\mathcal{M}$ is Quillen equivalent to modules over $\mathcal{O}_\mathcal{G}$. So going from one generator to many replaces a ring spectrum by a spectral category, and stable model categories with compact generators are the same thing as module categories over spectral categories.

15 Bar Constructions and Derived Tensor Products

Construction 15.1 (Two-sided bar construction)**.** Let M be a right R-module and N a left R-module. The simplicial spectrum $B_\bullet(M, R, N)$ has

$$B_q(M, R, N) = M \wedge R^{\wedge q} \wedge N.$$

The first and last face maps use the actions on M and N, the inner face maps multiply adjacent copies of R, and the degeneracies insert the unit $\mathbb{S} \to R$. Its geometric realization is

$$B(M, R, N) = |B_\bullet(M, R, N)|.$$

The first few terms are

$$M \wedge N \;\leftleftarrows\; M \wedge R \wedge N \;\lefttriplearrows\; M \wedge R \wedge R \wedge N \quad \cdots.$$

The coequalizer of the first two arrows is $M \wedge_R N$ by definition, so there is a map

$$B(M, R, N) \longrightarrow M \wedge_R N.$$

The realization of the whole simplicial object is the homotopy-correct version of that coequalizer, and (for cofibrant R, M, and N)

$$B(M, R, N) \simeq M \wedge^{\mathbf{L}}_R N.$$

This is the analogue of computing Tor with the bar resolution.

Definition 15.2 (Cyclic bar construction)**.** For an associative ring spectrum R, the cyclic bar construction is the simplicial spectrum with

$$B^{\mathrm{cy}}_q(R) = R^{\wedge(q+1)}.$$

The faces multiply adjacent factors, except for the last one, which brings the last factor around to the front and multiplies it with the first. The degeneracies insert units.

It is the bar construction with the two ends glued together. Its realization is topological Hochschild homology,

$$\mathrm{THH}(R) \simeq |B^{\mathrm{cy}}_\bullet(R)|,$$

and the cyclic permutations of the factors give $\mathrm{THH}(R)$ an action of the circle. With coefficients in an R-bimodule M one replaces one of the factors by M. Day 4 continues with THH.

16 Operads and Coherent Algebra

16.1 Loop-space recognition

Operads come from the question of when a space X is a loop space, or an n-fold loop space,

$$X \simeq \Omega^n Y?$$

A loop space has a multiplication by concatenating loops, which is associative up to homotopy, and the homotopies satisfy further relations up to higher homotopy. The answer will be that X is an n-fold loop space when it has all of this structure, organized by an E_n-operad, and is grouplike.

Example 16.1 (Unit octonions)**.** A multiplication with unit is not enough. The unit octonions make S^7 an H-space, with unit and inverses, but the multiplication is not homotopy associative and S^7 is not a loop space.

16.2 Endomorphism operads

Let $(\mathcal{V}, \otimes, \mathbf{1})$ be a closed symmetric monoidal category. For an object X, collect all the operations with n inputs and one output.

Definition 16.2 (Endomorphism operad)**.** The *endomorphism operad* of X is the sequence of objects

$$\mathrm{End}_X(n) = \underline{\mathrm{Hom}}_{\mathcal{V}}(X^{\otimes n}, X), \qquad n \geq 0.$$

The symmetric group Σ_n acts on $\mathrm{End}_X(n)$ by permuting inputs, the identity of X is a unit $\mathbf{1} \to \mathrm{End}_X(1)$, and operations can be substituted into one another:

$$\mathrm{End}_X(k) \otimes \mathrm{End}_X(n_1) \otimes \cdots \otimes \mathrm{End}_X(n_k) \longrightarrow \mathrm{End}_X(n_1 + \cdots + n_k).$$

For spaces $\mathrm{End}_X(n) = \mathrm{Map}(X^n, X)$, and for spectra it is the function spectrum $F(X^{\wedge n}, X)$.

16.3 Symmetric sequences and the substitution product

An operad is an abstract version of this.

Definition 16.3 (Symmetric sequence)**.** A *symmetric sequence* in $\mathcal{V}$ is a sequence of objects

$$A(0), A(1), A(2), \dots$$

where $A(n)$ has a right action of Σ_n. Morphisms are levelwise equivariant maps.

Besides the tensor product from Day 2, symmetric sequences have a second, non-symmetric product, the substitution product

$$(A \circ B)(n) = \coprod_{k \geq 0} A(k) \otimes_{\Sigma_k} \left[\coprod_{n_1 + \cdots + n_k = n} \big(B(n_1) \otimes \cdots \otimes B(n_k)\big) \otimes_{\Sigma_{n_1} \times \cdots \times \Sigma_{n_k}} \Sigma_n \right].$$

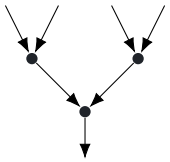
The formula looks worse than it is: a point of $(A \circ B)(n)$ is a k-ary operation from A with operations from B of arities $n_1, \dots, n_k$ plugged into its inputs.

Definition 16.4 (Operad)**.** An *operad* $\mathcal{O}$ is a monoid for the substitution product. Unwinding, it consists of objects $\mathcal{O}(n)$ with Σ_n-actions, a unit $\mathbf{1} \to \mathcal{O}(1)$, and equivariant composition maps

$$\gamma \colon \mathcal{O}(k) \otimes \mathcal{O}(n_1) \otimes \cdots \otimes \mathcal{O}(n_k) \longrightarrow \mathcal{O}(n_1 + \cdots + n_k)$$

that are associative and unital.

The usual picture is a tree, with operations at the nodes:



$\rightsquigarrow$ one four-input operation.

Composition contracts the internal edges, and associativity says that for a taller tree the order of contraction does not matter.

16.4 Algebras over an operad

Definition 16.5 ($\mathcal{O}$-algebra)**.** An *algebra over* $\mathcal{O}$ is an object $X \in \mathcal{V}$ with a map of operads

$$\mathcal{O} \longrightarrow \mathrm{End}_X,$$

or equivalently with compatible action maps

$$\mathcal{O}(n) \otimes_{\Sigma_n} X^{\otimes n} \longrightarrow X, \qquad n \geq 0.$$

One should think of $\mathcal{O}(n)$ as the n-ary operations that every $\mathcal{O}$-algebra has.

17 Examples of Operads and Recognition

17.1 Strict associative and commutative operads

Example 17.1 (The commutative operad)**.** Let

$$\mathrm{Comm}(n) = \mathbf{1}$$

for all n, with trivial Σ_n-action. There is exactly one n-ary operation for each n, and equivariance forces the binary one to satisfy

$$\mu\tau = \mu,$$

so a Comm-algebra is a commutative monoid in $\mathcal{V}$. In $\mathcal{V} = \mathbf{Sp}$ these are the strictly commutative ring spectra.

Example 17.2 (The associative operad)**.** Let

$$\mathrm{Ass}(n) = \coprod_{\sigma \in \Sigma_n} \mathbf{1},$$

with Σ_n permuting the summands. There is one n-ary operation for each ordering of the inputs, and the algebras are the associative monoids.

17.2 A_n, A_∞, and little cubes

The homotopy-coherent versions are obtained by replacing the points of $\mathrm{Ass}(n)$ and $\mathrm{Comm}(n)$ with contractible spaces. For associativity this goes back to Stasheff: the associahedra parametrize the ways of bracketing a product, and an A_∞-space has a multiplication together with a map from each associahedron. An A_n-space has these maps only up to the nth associahedron.

Definition 17.3 (Little n-cubes operad)**.** Let $\mathcal{C}_n(k)$ be the space of k-tuples of rectilinear embeddings

$$(0,1)^n \hookrightarrow (0,1)^n$$

with pairwise disjoint images. The group Σ_k permutes the labels of the little cubes, and composition inserts a configuration of little cubes into each little cube of another configuration. This is the *little n-cubes operad*. An *E_n-operad* is an operad weakly equivalent to $\mathcal{C}_n$, and an *E_n-algebra* is an algebra over one.

For $n = 1$ the little cubes are intervals in a line, $\mathcal{C}_1(k)$ is homotopy equivalent to Σ_k, and E_1 is the same as A_∞. As n grows there is more room to move the cubes past each other, so the multiplication becomes more commutative. In the limit, an E_∞-operad has contractible spaces of operations with free Σ_k-actions.

An n-fold loop space $\Omega^n Y$ is a $\mathcal{C}_n$-algebra: given k little cubes and k maps $S^n \to Y$, run the ith map on the ith cube and send everything else to the basepoint. The recognition principle says that this is essentially the only source of examples.

Theorem 17.4 (May recognition principle)**.** *Let X be a grouplike algebra over the little n-cubes operad. Then there is a space $B^n X$ with*

$$X \simeq \Omega^n B^n X.$$

For $n = \infty$, a grouplike E_∞-space is the zeroth space of a connective spectrum [May72].

Grouplike means that the monoid $\pi_0 X$ is a group.

18 Γ-Spaces and Infinite Loop Spaces

18.1 Special Γ-spaces

Segal's Γ-spaces from Day 2 are a second way to encode an E_∞-structure, and they do it without choosing an operad. Recall from Definition 9.1 that a Γ-space X is special if the Segal maps

$$\varphi_n \colon X(n_+) \longrightarrow X(1_+)^n$$

are weak equivalences. The multiplication on $X(1_+)$ comes from the fold map

$$\nabla \colon 2_+ \longrightarrow 1_+$$

by inverting a Segal equivalence:

$$X(1_+) \times X(1_+) \xleftarrow{\simeq} X(2_+) \xrightarrow{X(\nabla)} X(1_+).$$

It is only defined up to homotopy, and the spaces $X(n_+)$ for larger n carry the coherence. A very special Γ-space is one for which this multiplication makes $\pi_0 X(1_+)$ a group, which is the grouplike condition of the recognition principle.

18.2 The Segal machine

The Segal machine is the passage from X to its spectrum on Day 2: extend X to pointed spaces and put

$$E_n = X(S^n),$$

with structure maps

$$\Sigma E_n \longrightarrow E_{n+1}.$$

If X is very special, the adjoints

$$E_n \longrightarrow \Omega E_{n+1}$$

are weak equivalences, so $E = (E_0, E_1, \ldots)$ is an Ω-spectrum with

$$X(1_+) \simeq \Omega^\infty E.$$

If X is only special, this holds from E_1 on, and $E_0 \to \Omega E_1$ is a group completion.

So there are now two infinite loop space machines, May's operadic one and Segal's. The May–Thomason theorem says that any two machines satisfying a short list of axioms give equivalent spectra.

19 Rectification and Its Limits

Sometimes a homotopy-coherent structure can be replaced by a strict one. The classical case is the Moore loop space

$$\Omega^{\mathrm{M}} X = \{(\ell, \gamma) \mid \ell \geq 0,\ \gamma \colon [0, \ell] \to X,\ \gamma(0) = \gamma(\ell) = *\},$$

whose points are loops with a length. Concatenating two loops adds their lengths and needs no reparametrization, so it is strictly associative:

$$((\ell_1, \gamma_1) \cdot (\ell_2, \gamma_2)) \cdot (\ell_3, \gamma_3) = (\ell_1, \gamma_1) \cdot ((\ell_2, \gamma_2) \cdot (\ell_3, \gamma_3)).$$

For well-pointed X the Moore loop space is homotopy equivalent to ΩX, so every loop space is equivalent to a topological monoid.

In general, a weak equivalence of operads

$$\mathcal{O} \longrightarrow \mathcal{P}$$

induces an adjunction between $\mathcal{O}$-algebras and $\mathcal{P}$-algebras, and under cofibrancy conditions on the operads this is a Quillen equivalence. Applied to $A_\infty \to \mathrm{Ass}$ it says that A_∞-algebras can be rectified to strictly associative ones.

Warning 19.1 (Rectification depends on the context). Whether rectification is possible depends on the operad and on the category. For $E_\infty \to \mathrm{Comm}$ in spaces it fails, by the fact about strictly commutative topological monoids behind Lewis's obstruction. In spectra with the positive model structure it holds, and E_∞ ring spectra are equivalent to commutative ring spectra.

20 Eilenberg–Mac Lane Spectra and Derived Algebra

20.1 From chain complexes to module spectra

The last topic of the day connects ring spectra with ordinary homological algebra. The starting point is the Dold–Kan correspondence

$$N \colon s\,\mathrm{Mod}_A \rightleftarrows \mathbf{Ch}_{\geq 0}(A) : \Gamma,$$

an equivalence of categories between simplicial A-modules and nonnegatively graded chain complexes. Stabilizing both sides relates HA-modules to unbounded chain complexes.

Theorem 20.1 (Modules over HA)**.** *For a ring A, the model category of HA-module spectra is Quillen equivalent (by a zigzag) to unbounded chain complexes of A-modules. In particular*

$$\mathrm{Ho}(HA\text{-}\mathrm{Mod}) \simeq D(A)$$

as triangulated categories.

Theorem 20.2 (Shipley)**.** *The homotopy theory of $H\mathbb{Z}$-algebra spectra is equivalent, by a zigzag of Quillen equivalences, to the homotopy theory of differential graded algebras over $\mathbb{Z}$.*

This is about associative algebras. For commutative algebras the analogous statement is true over $\mathbb{Q}$ and false in general: commutative $H\mathbb{F}_p$-algebras have Dyer–Lashof operations, which commutative differential graded algebras do not see.

20.2 Algebraic and nonalgebraic stable homotopy theories

Definition 20.3 (Algebraic stable homotopy theory)**.** Roughly, a stable homotopy theory is *algebraic* if it can be modeled by differential graded categories, or equivalently if its mapping spectra are $H\mathbb{Z}$-modules compatibly with composition. Otherwise it is *nonalgebraic*.

Derived categories are algebraic. Spectra are not: the endomorphism spectrum of the unit is

$$\underline{\mathrm{End}}_{\mathbf{Sp}}(\mathbb{S}) \simeq \mathbb{S},$$

and $\mathbb{S}$ is not an $H\mathbb{Z}$-algebra.

The graded ring $\pi_* R$ does not determine a ring spectrum R either: two ring spectra with the same homotopy ring can differ in their k-invariants and higher operations.

Part IV

Day 4: Rings and Modules, Part II

Day 4 was a tour of the invariants one can attach to a ring spectrum R: Thom spectra built from its units, THH, the cotangent complex, algebraic K-theory, and the Picard and Brauer spaces. From here on everything is derived: colimits, smash products, and mapping objects mean the homotopy invariant ones, unless a point-set model is mentioned explicitly. The ∞-categorical background is in [Lur17].

21 Thom Spaces and the Stable J-Homomorphism

21.1 Fiberwise compactification

Thom spaces already appeared on Day 2 as the mapping spaces of the indexing category for orthogonal spectra. Here is the definition again in the form we need. Let $p\colon E \to X$ be a real vector bundle of rank n, and let $E^+ \to X$ be its fiberwise one-point compactification. The fiber over x is

$$(E_x)^+ \cong (\mathbb{R}^n)^+ \cong S^n,$$

and the points at infinity form a section

$$s_\infty\colon X \longrightarrow E^+.$$

Definition 21.1 (Thom space)**.** The *Thom space* of E is the quotient

$$\mathrm{Th}(E) = E^+/s_\infty(X).$$

After choosing a fiberwise inner product this is the disk bundle modulo the sphere bundle,

$$\mathrm{Th}(E) \cong D(E)/S(E).$$

For the trivial bundle $\varepsilon^n = X \times \mathbb{R}^n$ we get

$$\mathrm{Th}(\varepsilon^n) \cong X_+ \wedge S^n \cong \Sigma^n X_+,$$

so the Thom space of a general bundle is a twisted suspension of X_+. The construction turns sums into smash products: for $E \to X$ and $F \to Y$,

$$\mathrm{Th}(E \boxplus F) \cong \mathrm{Th}(E) \wedge \mathrm{Th}(F),$$

where $E \boxplus F$ is the external direct sum over $X \times Y$.

A rank-n bundle is classified by a map

$$c_E\colon X \longrightarrow BO(n),$$

with $E \cong c_E^*\gamma_n$ for the universal bundle γ_n. Adding a trivial line corresponds to the inclusion $BO(n) \to BO(n+1)$, which is covered by a bundle map

$$\gamma_n \oplus \varepsilon^1 \longrightarrow \gamma_{n+1}.$$

Taking Thom spaces gives

$$\Sigma\,\mathrm{Th}(\gamma_n) \cong \mathrm{Th}(\gamma_n \oplus \varepsilon^1) \longrightarrow \mathrm{Th}(\gamma_{n+1}),$$

and these are the structure maps of the spectrum MO. The same recipe with other families of classifying spaces gives MSO, $MSpin$, and MU.

21.2 Spherical fibrations and stabilization

The Thom space only uses the fiberwise compactification E^+, which is a bundle of spheres, and only up to fiberwise homotopy equivalence. Let

$$G(n) = \mathrm{hAut}_*(S^n)$$

be the monoid of based self-homotopy equivalences of S^n. One-point compactification gives a map

$$O(n) \longrightarrow G(n),$$

and so a map of classifying spaces

$$BO(n) \longrightarrow BG(n).$$

These are compatible with adding a trivial line on both sides, and the colimit over n is the stable J-homomorphism

$$BO \xrightarrow{\quad J \quad} BGL_1(\mathbb{S}).$$

Here $GL_1(\mathbb{S})$ is the space of units of the sphere spectrum, defined in the next section. On homotopy groups J is the classical J-homomorphism $\pi_k O \to \pi_k \mathbb{S}$, but J as a map of spaces remembers more than its effect on homotopy groups.

A stable virtual bundle on X is a map

$$\xi \colon X \longrightarrow BO \times \mathbb{Z}.$$

Composing with J turns it into a family of spectra parametrized by X, each equivalent to a sphere, and the Thom spectrum of ξ is the colimit of this family. The next section takes this as the definition.

22 Units, Lines, and Generalized Thom Spectra

22.1 Sphere lines, R-lines, and the Thom colimit

An $\mathbb{S}$-*line* is a spectrum equivalent to $\mathbb{S}$. The $\mathbb{S}$-lines and the equivalences between them form an ∞-groupoid, which is connected, so

$$\mathbb{S}\text{-line} \simeq B\,\mathrm{Aut}_{\mathbf{Sp}}(\mathbb{S}) \simeq BGL_1(\mathbb{S}).$$

The same works over any E_∞ ring spectrum R. An R-*line* is an R-module equivalent to R, and R-line is the ∞-groupoid of R-lines and equivalences. Since

$$\mathrm{Aut}_R(R) \simeq GL_1(R),$$

we get

$$R\text{-line} \simeq BGL_1(R).$$

A map

$$f \colon X \longrightarrow R\text{-line} \simeq BGL_1(R)$$

is a local system of free rank-one R-modules on X.

Definition 22.1 (Generalized Thom spectrum)**.** Regard f as a functor $X \to \mathrm{Mod}_R$. Its *Thom spectrum* is the R-module

$$M_R f = \mathrm{colim}\big(X \xrightarrow{f} R\text{-line} \hookrightarrow \mathrm{Mod}_R\big).$$

For $R = \mathbb{S}$ and $f = J \circ \xi$ this is the classical Thom spectrum of ξ. If f is nullhomotopic the local system is constant, and the colimit is

$$M_R f \simeq R \wedge \Sigma^\infty_+ X,$$

which is the analogue of the trivial bundle.

22.2 Units and the assembly adjunction

Now for the units. The space $\Omega^\infty R$ is an E_∞-monoid under multiplication, and its monoid of components is the multiplicative monoid of $\pi_0 R$. Let $GL_1(R)$ be the union of the components that are units:

$$GL_1(R) \subseteq \Omega^\infty R, \qquad \pi_0 GL_1(R) = (\pi_0 R)^\times.$$

This is a grouplike E_∞-space, so by the recognition principle it is the infinite loop space of a connective spectrum $gl_1(R)$,

$$\Omega^\infty gl_1(R) \simeq GL_1(R).$$

Each component of $GL_1(R)$ is a component of $\Omega^\infty R$, so

$$\pi_i gl_1(R) \cong \pi_i R \qquad (i > 0).$$

As spectra $gl_1(R)$ and R are very different, because one infinite loop structure comes from multiplication and the other from addition.

The functor GL_1 is a right adjoint,

$$\Sigma^\infty_+ \colon \mathrm{CGrp}(\mathbf{Top}) \rightleftarrows \mathrm{CAlg}(\mathbf{Sp}) : GL_1,$$

where the category on the left is that of grouplike E_∞-spaces. The counit at R is the assembly map

$$\Sigma^\infty_+ GL_1(R) \longrightarrow R.$$

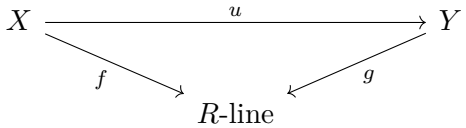
For an ordinary ring A this is the homomorphism

$$\mathbb{Z}[A^\times] \longrightarrow A$$

from the group ring of the units.

22.3 Functoriality of the Thom construction

Being a colimit, the Thom spectrum is functorial in the base. A triangle



together with a homotopy $f \simeq g \circ u$ induces a map

$$M_R f \longrightarrow M_R g.$$

22.4 Multiplicative Thom spectra

Given $f\colon X \to R$-line and $g\colon Y \to R$-line, the external product $f \boxtimes g$ is

$$X \times Y \xrightarrow{f\times g} R\text{-line} \times R\text{-line} \xrightarrow{\otimes_R} R\text{-line},$$

and the Thom spectrum functor is symmetric monoidal:

$$M_R(f \boxtimes g) \simeq M_R f \wedge_R M_R g.$$

So if X is an E_n-space and f is an E_n-map, then $M_R f$ is an E_n R-algebra. For $n = \infty$ it is a commutative R-algebra. This is where the ring structures on MO and MU come from.

23 Topological Hochschild Homology

23.1 Simplicial and cyclic objects

Recall that the realization of a simplicial object $X_\bullet$ is

$$|X_\bullet| = \operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} X_n.$$

A *cyclic object* is a simplicial object with an extra operator t_n of order $n+1$ in each degree, compatible with faces and degeneracies. The main example is the cyclic bar construction from Day 3, now relative to a commutative ring spectrum R: for an R-algebra A,

$$B^{\mathrm{cy}}_n(A/R) = A^{\wedge_R(n+1)}, \qquad n \geq 0,$$

with faces multiplying adjacent factors, degeneracies inserting the unit, and t_n rotating the factors.

Definition 23.1 (Topological Hochschild homology)**.** For an E_1 R-algebra A, the relative topological Hochschild homology is

$$\mathrm{THH}^R(A) = |B^{\mathrm{cy}}_\bullet(A/R)|\,.$$

We write $\mathrm{THH}(A) = \mathrm{THH}^{\mathbb{S}}(A)$.

It can also be written as a derived tensor product over the enveloping algebra $A^e = A \wedge_R A^{\mathrm{op}}$,

$$\mathrm{THH}^R(A) \simeq A \wedge^{\mathbf{L}}_{A\wedge_R A^{\mathrm{op}}} A,$$

which is the usual definition of Hochschild homology. Also, $\mathrm{THH}^R(R) \simeq R$, and if A is commutative then $\mathrm{THH}^R(A)$ is a commutative A-algebra.

23.2 The circle action

The realization of a cyclic object has an action of the circle, so

$$S^1 \curvearrowright \mathrm{THH}^R(A).$$

Warning 23.2 (The circle action is not yet cyclotomic structure)*.* Topological cyclic homology needs more than the circle action. It also uses the cyclotomic Frobenius maps on THH, which are extra structure:

$$\text{cyclic object} \implies S^1\text{-spectrum}, \qquad \text{cyclotomic spectrum} \implies \mathrm{TC}\,.$$

23.3 Factorization homology and tensoring over spaces

There are two more descriptions of THH. For an E_1-algebra A it is factorization homology over the circle,

$$\int_{S^1} A \simeq \mathrm{THH}(A).$$

For commutative A there is something simpler. The category CAlg_R is tensored over spaces, with $X \otimes A$ defined by

$$\mathrm{Map}_{\mathrm{CAlg}_R}(X \otimes A, B) \simeq \mathrm{Map}_{\mathbf{Top}}\Big(X, \mathrm{Map}_{\mathrm{CAlg}_R}(A, B)\Big),$$

and

$$\mathrm{THH}^R(A) \simeq S^1 \otimes A \qquad (A \in \mathrm{CAlg}_R).$$

In this form the circle action is visible, since S^1 acts on itself. A basepoint of S^1 and the map $S^1 \to *$ give maps

$$A \longrightarrow S^1 \otimes A \longrightarrow A$$

whose composite is the identity.

24 TAQ and the Cotangent Complex

24.1 Tangent categories and linearization

The cotangent complex is defined by stabilization. For an ∞-category $\mathcal{D}$ and an object B, the tangent category at B is

$$T_B\mathcal{D} = \mathbf{Sp}(\mathcal{D}_{/B}),$$

and the cotangent object is the suspension spectrum of the identity of B,

$$L_B = \Sigma^\infty_{B,+}(\mathrm{id}_B) \in T_B\mathcal{D}.$$

As B varies these assemble into a functor

$$L\colon \mathcal{D} \longrightarrow T\mathcal{D}, \qquad B \longmapsto (B, L_B).$$

For $\mathcal{D} = \mathrm{CAlg}_R$ the tangent category at A is the category of A-modules,

$$T_A\,\mathrm{CAlg}_R \simeq \mathrm{Mod}_A,$$

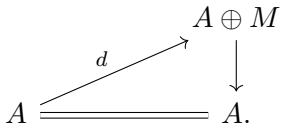
so the cotangent object of A is an A-module, written $L_{A/R}$.

24.2 Cotangent specialization, square-zero extensions, and derivations

Here is what that means concretely. Fix a map of E_∞ rings $R \to A$ and an A-module M. The split square-zero extension $A \oplus M$ is the commutative R-algebra over A whose underlying module is $A \oplus M$ and whose multiplication is, informally,

$$(a, m)(a', m') = (aa', am' + a'm), \qquad mm' = 0.$$

A derivation of A over R with values in M is a section of the projection $A \oplus M \to A$ in commutative R-algebras:



Let $\mathrm{Der}_R(A, M)$ be the space of such sections.

Definition 24.1 (Cotangent complex and TAQ)**.** The relative cotangent complex $L_{A/R} \in \mathrm{Mod}_A$ is the A-module representing derivations:

$$\mathrm{Map}_{\mathrm{Mod}_A}(L_{A/R}, M) \simeq \mathrm{Der}_R(A, M).$$

It is also called topological André–Quillen homology and written

$$\mathrm{TAQ}^R(A) = L_{A/R}.$$

With coefficients in M,

$$\mathrm{TAQ}^R(A; M) = M \wedge_A L_{A/R}, \qquad \mathrm{TAQ}_R(A; M) = \underline{\mathrm{Map}}_A(L_{A/R}, M).$$

So $\mathrm{Der}_R(A, M) \simeq \Omega^\infty \mathrm{TAQ}_R(A; M)$. Two properties carry over from the algebraic cotangent complex. For maps $R \to A \to B$ there is a cofiber sequence of B-modules

$$B \wedge_A L_{A/R} \longrightarrow L_{B/R} \longrightarrow L_{B/A},$$

and for a free commutative algebra $A = \mathbb{P}_R(N)$,

$$L_{A/R} \simeq A \wedge_R N.$$

24.3 Nonsplit extensions and Postnikov invariants

Derivations with values in a shift of M classify square-zero extensions that do not split. A map

$$\eta \colon L_{A/R} \longrightarrow \Sigma M$$

is a derivation $A \to A \oplus \Sigma M$, and we define A^η as the pullback of it along the zero derivation:

$$\begin{array}{ccc} A^\eta & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle \eta} \\ A & \xrightarrow[0]{} & A \oplus \Sigma M. \end{array}$$

The fiber of $A^\eta \to A$ is M.

The main example is the Postnikov tower of a connective commutative ring spectrum A,

$$\cdots \longrightarrow \tau_{\leq n} A \longrightarrow \tau_{\leq n-1} A \longrightarrow \cdots \longrightarrow \tau_{\leq 0} A.$$

Each map $\tau_{\leq n} A \to \tau_{\leq n-1} A$ is a square-zero extension by $\Sigma^n H\pi_n A$, and the derivation that classifies it is the nth k-invariant of A as a commutative ring spectrum.

25 Algebraic K-Theory of Stable Categories

25.1 The Grothendieck group

Let $\mathcal{C}$ be a small stable ∞-category. Its Grothendieck group $K_0(\mathcal{C})$ is generated by symbols $[X]$, one for each object, with a relation

$$[Y] = [X] + [Z]$$

for every cofiber sequence

$$X \longrightarrow Y \longrightarrow Z.$$

It follows that

$$[0] = 0, \qquad [X \oplus Y] = [X] + [Y], \qquad [\Sigma X] = -[X].$$

For a ring spectrum R we apply this to the compact modules, also called perfect modules,

$$\mathrm{Perf}(R) = (\mathrm{Mod}_R)^{\omega}, \qquad K_0(R) = K_0(\mathrm{Perf}(R)).$$

We use perfect modules because all of Mod_R would give zero, by an Eilenberg swindle.

Thomason's theorem is a nice illustration of what K_0 sees. In a small triangulated category $\mathcal{T}$, call a full triangulated subcategory $\mathcal{A}$ *dense* if every object of $\mathcal{T}$ is a direct summand of an object of $\mathcal{A}$. Then

$$\left\{ \begin{array}{c} \text{strictly full dense triangulated} \\ \text{subcategories } \mathcal{A} \subseteq \mathcal{T} \end{array} \right\} \longleftrightarrow \{\text{subgroups of } K_0(\mathcal{T})\},$$

where $\mathcal{A}$ corresponds to the image of $K_0(\mathcal{A}) \to K_0(\mathcal{T})$.

26 Universal Invariants and Noncommutative Motives

26.1 Morita invariance, additivity, and localization

Let $\mathbf{Cat}_\infty^{\mathrm{perf}}$ be the ∞-category of small idempotent-complete stable ∞-categories and exact functors. Many invariants of rings are really functors on $\mathbf{Cat}_\infty^{\mathrm{perf}}$, evaluated on $\mathrm{Perf}(R)$.

Definition 26.1 (Additive and localizing invariants)**.** Let $E\colon \mathbf{Cat}_\infty^{\mathrm{perf}} \to \mathcal{D}$ be a functor to a stable presentable ∞-category that inverts Morita equivalences and preserves filtered colimits.

1. E is an *additive invariant* if it sends every split-exact sequence
$$\mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C}$$
to a direct sum decomposition $E(\mathcal{B}) \simeq E(\mathcal{A}) \oplus E(\mathcal{C})$.
2. E is a *localizing invariant* if it sends every exact (Verdier) sequence to a cofiber sequence
$$E(\mathcal{A}) \longrightarrow E(\mathcal{B}) \longrightarrow E(\mathcal{C}).$$

Every localizing invariant is additive. There are universal invariants of both kinds,

$$\mathbf{Cat}_\infty^{\mathrm{perf}} \xrightarrow{U_{\mathrm{add}}} \mathrm{Mot}_{\mathrm{add}}, \qquad\qquad \mathbf{Cat}_\infty^{\mathrm{perf}} \xrightarrow{U_{\mathrm{loc}}} \mathrm{Mot}_{\mathrm{loc}},$$

whose targets are the categories of noncommutative motives. An additive invariant is the same thing as a colimit-preserving functor out of $\mathrm{Mot}_{\mathrm{add}}$, and likewise for localizing invariants and $\mathrm{Mot}_{\mathrm{loc}}$. Connective K-theory is additive, nonconnective K-theory is localizing, and the last subsection says in what sense they are the universal examples.

26.2 Waldhausen categories and the $S_\bullet$-construction

The K-theory spectrum is built by Waldhausen's $S_\bullet$-construction. Waldhausen's original input is a category with cofibrations and weak equivalences. A stable ∞-category fits in by declaring every map to be a cofibration.

Let $\mathrm{Ar}[n]$ be the category of arrows in $[n] = \{0 < 1 < \cdots < n\}$, that is, of pairs $i \leq j$. An object of $S_n\mathcal{C}$ is a diagram

$$X\colon \mathrm{Ar}[n] \longrightarrow \mathcal{C}$$

such that

$$X_{ii} \simeq 0$$

and every square

$$\begin{array}{ccc} X_{ij} & \longrightarrow & X_{ik} \\ \downarrow & & \downarrow \\ 0 \simeq X_{jj} & \longrightarrow & X_{jk} \end{array} \qquad (i \leq j \leq k)$$

is a pushout. The top row

$$0 = X_{00} \longrightarrow X_{01} \longrightarrow \cdots \longrightarrow X_{0n}$$

is a filtered object with n steps, and the rest of the diagram is the collection of its subquotients $X_{ij} = X_{0j}/X_{0i}$. So $S_n\mathcal{C}$ is the category of n-step filtrations with chosen subquotients.

The K-theory space is

$$\Omega\,|(S_\bullet\mathcal{C})^{\simeq}|\,,$$

and iterating the construction gives the connective spectrum $K(\mathcal{C})$, with

$$\pi_0 K(\mathcal{C}) \cong K_0(\mathcal{C}).$$

26.3 THH of stable categories and the cyclotomic trace

THH also extends from ring spectra to categories. For a small spectral category $\mathcal{C}$, the cyclic bar construction has

$$\bigvee_{x_0,\ldots,x_n} \mathcal{C}(x_0,x_1) \wedge \mathcal{C}(x_1,x_2) \wedge \cdots \wedge \mathcal{C}(x_n,x_0)$$

in degree n, and its realization is $\mathrm{THH}(\mathcal{C})$. This is Morita invariant, and $\mathrm{THH}(\mathrm{Perf}(A)) \simeq \mathrm{THH}(A)$.

K-theory and THH are related by the Dennis trace, a natural transformation

$$K(\mathcal{C}) \longrightarrow \mathrm{THH}(\mathcal{C}).$$

It factors through topological cyclic homology, and the lift

$$K(\mathcal{C}) \longrightarrow \mathrm{TC}(\mathcal{C})$$

is the cyclotomic trace.

26.4 Corepresenting K-theory

Let $\mathbb{S}^\omega = \mathrm{Perf}(\mathbb{S})$ be the category of finite spectra. Its motive is the unit of $\mathrm{Mot}_{\mathrm{add}}$, and maps out of it compute K-theory:

$$\mathrm{Map}_{\mathrm{Mot}_{\mathrm{add}}}(U_{\mathrm{add}}(\mathbb{S}^\omega), U_{\mathrm{add}}(\mathcal{A})) \simeq K(\mathcal{A}).$$

The same formula in $\mathrm{Mot}_{\mathrm{loc}}$ gives nonconnective K-theory. More generally, for smooth and proper $\mathcal{A}$,

$$\mathrm{Map}_{\mathrm{Mot}}(U(\mathcal{A}), U(\mathcal{B})) \simeq K(\mathcal{A}^{\mathrm{op}} \otimes \mathcal{B}).$$

27 Picard Spaces and Generalized Thom Spectra

27.1 Invertible objects

Let $\mathcal{C}$ be a symmetric monoidal ∞-category. An object L is *invertible* if there is an object L^{-1} with

$$L \otimes L^{-1} \simeq \mathbf{1}.$$

The *Picard space* is the ∞-groupoid of invertible objects and equivalences between them,

$$\mathrm{Pic}(\mathcal{C}) = (\mathcal{C}^{\simeq})^{\times}.$$

The tensor product makes it a grouplike E_∞-space, so it is the infinite loop space of a connective spectrum $\mathrm{pic}(\mathcal{C})$.

For an E_∞ ring R we write

$$\mathrm{Pic}(R) = \mathrm{Pic}(\mathrm{Mod}_R).$$

The loop space at the unit R is the space of automorphisms of R,

$$\Omega_R \,\mathrm{Pic}(R) \simeq \mathrm{Aut}_R(R) \simeq GL_1(R),$$

so the homotopy groups are

$$\begin{aligned} \pi_0 \,\mathrm{pic}(R) &= \mathrm{Pic}(R), \\ \pi_1 \,\mathrm{pic}(R) &= (\pi_0 R)^{\times}, \\ \pi_i \,\mathrm{pic}(R) &= \pi_{i-1} R, \qquad i \geq 2, \end{aligned}$$

where $\mathrm{Pic}(R)$ is the group of equivalence classes of invertible R-modules.

27.2 Descent

The functor $R \mapsto \mathrm{Pic}(R)$ satisfies étale descent, and this is the main tool for computing Picard groups. For connective R the descent spectral sequence starts from the étale cohomology of the homotopy sheaves of pic.

27.3 Picard-valued Thom twists

The component of $\mathrm{Pic}(R)$ containing R is the space of R-lines from the section on units,

$$\mathrm{Pic}(R)_{[R]} \simeq R\text{-line} \simeq BGL_1(R).$$

The Thom spectrum construction works just as well for a map into all of $\mathrm{Pic}(R)$,

$$f\colon X \longrightarrow \mathrm{Pic}(R), \qquad M_R f = \mathrm{colim}(X \xrightarrow{f} \mathrm{Pic}(R) \hookrightarrow \mathrm{Mod}_R),$$

and now the fibers are allowed to be shifts of R or other invertible modules.

27.4 The complex J-homomorphism and MUP

Complex bordism gives an example. The complex J-homomorphism extends over the components recording the virtual rank,

$$J_{\mathbb{C}} \colon BU \times \mathbb{Z} \longrightarrow \mathrm{Pic}(\mathbb{S}),$$

with the component of rank n mapping to lines equivalent to S^{2n}. Its Thom spectrum is periodic complex bordism MUP. As a spectrum

$$MUP \simeq \bigvee_{n \in \mathbb{Z}} \Sigma^{2n} MU,$$

and the multiplication sends the summands with indices m and n to the summand with index $m + n$.

28 Brauer Spaces and Morita-Invertible Categories

28.1 Morita-invertible module categories

One level up, we look at invertible objects among R-linear categories. An associative R-algebra A is *Azumaya* if Mod_A is invertible as an R-linear category. Equivalently, A is a compact generator of Mod_R and the map

$$A \wedge_R A^{\mathrm{op}} \longrightarrow \underline{\mathrm{End}}_R(A)$$

is an equivalence. The *Brauer space* $\mathrm{Br}(R)$ is the ∞-groupoid of invertible R-linear categories, and its components are the Morita classes of Azumaya algebras.

28.2 Étale descent and low homotopy groups

The Brauer space also satisfies étale descent. For a connective E_∞ ring R its homotopy groups are

$$\begin{aligned}
&\pi_0 \mathrm{Br}(R) \cong H^1_{\text{ét}}(\mathrm{Spec}\, \pi_0 R; \mathbb{Z}) \oplus H^2_{\text{ét}}(\mathrm{Spec}\, \pi_0 R; \mathbb{G}_m),\\
&\pi_1 \mathrm{Br}(R) \cong H^0_{\text{ét}}(\mathrm{Spec}\, \pi_0 R; \mathbb{Z}) \oplus H^1_{\text{ét}}(\mathrm{Spec}\, \pi_0 R; \mathbb{G}_m),\\
&\pi_2 \mathrm{Br}(R) \cong (\pi_0 R)^\times,\\
&\pi_k \mathrm{Br}(R) \cong \pi_{k-2} R, \qquad\qquad k \geq 3.
\end{aligned}$$

The second line is $\pi_0 \mathrm{Pic}(R)$, and the H^2 term in the first line is the cohomological Brauer group of $\pi_0 R$.

Understanding / Intuition

Units, Picard, and Brauer are the same idea at three categorical levels. A unit is an invertible endomorphism of R, a Picard object is an invertible R-module, and a Brauer object is an invertible category of modules. Taking loops goes down one level each time.

28.3 Looping and delooping relations

Precisely, with basepoint Mod_R,

$$\Omega_{\mathrm{Mod}_R} \mathrm{Br}(R) \simeq \mathrm{Pic}(R),$$

and looping once more gives

$$\Omega_R \operatorname{Pic}(R) \simeq GL_1(R).$$

In the other direction, delooping only recovers the component of the basepoint:

$$\operatorname{Pic}(R)_{[R]} \simeq BGL_1(R), \qquad \operatorname{Br}(R)_{[\operatorname{Mod}_R]} \simeq B\operatorname{Pic}(R).$$

29 Bipermutative Categories and ku

The day ended with the construction of ku as a ring spectrum. On Day 2 we saw that the topological groupoid $\mathcal{V}_{\mathbb{C}}$ of finite-dimensional complex inner-product spaces and unitary isomorphisms, with direct sum, gives the spectrum ku. The category has a second symmetric monoidal structure, the tensor product,

$$(\mathcal{V}_{\mathbb{C}}, \oplus, 0), \qquad (\mathcal{V}_{\mathbb{C}}, \otimes, \mathbb{C}),$$

and the two are related by distributivity isomorphisms

$$V \otimes (W \oplus U) \cong (V \otimes W) \oplus (V \otimes U)$$

and similarly in the other variable. After strictification this is a *bipermutative category*, the categorical version of a commutative semiring.

The additive structure gives the spectrum. The classifying space is

$$\coprod_{n \geq 0} BU(n),$$

and its group completion is

$$\mathbb{Z} \times BU \simeq \Omega^\infty ku.$$

The multiplicative structure then makes ku an E_∞ ring spectrum. Its homotopy ring is

$$\pi_* ku \cong \mathbb{Z}[u], \qquad |u| = 2,$$

with u the Bott class, and inverting u gives periodic complex K-theory,

$$KU \simeq ku[u^{-1}].$$

Part V

Day 5: Localization and Spectral Algebraic Geometry

The last day had two halves. The first was Bousfield localization: the general formalism, when localizations exist and when they are compatible with the smash product, and then the chromatic localizations of spectra. The second was a quick introduction to spectral algebraic geometry, ending with the description of TMF as global sections over the moduli of elliptic curves. As on Day 4, tensor products, mapping spaces, limits, and colimits are derived.

30 Bousfield Localization

30.1 Equivalences detected by a homology theory

Let E be a spectrum. A map $f\colon X \to Y$ is an *E-equivalence* if

$$E \wedge f\colon E \wedge X \longrightarrow E \wedge Y$$

is an equivalence, in other words if f is an isomorphism on E-homology. Since smashing with E preserves cofiber sequences, this is the same as

$$E \wedge \operatorname{cof}(f) \simeq 0.$$

For example, an $H\mathbb{Z}/2$-equivalence is a map inducing an isomorphism on mod-2 homology. Localizing at E means forcing the E-equivalences to be equivalences, which keeps the part of a spectrum that E can see. Two familiar constructions are built on this idea: the arithmetic fracture square, which splits a spectrum into its rational and p-primary parts, and Adams resolutions, which approximate a spectrum by objects built from E.

30.2 Local objects and local equivalences

The general setup needs only an ∞-category $\mathcal{C}$ and a collection S of maps that we would like to invert.

Definition 30.1 (Local object)**.** An object $Z \in \mathcal{C}$ is *S-local* if for every $s\colon A \to B$ in S, precomposition with s is an equivalence

$$s^*\colon \operatorname{Map}_{\mathcal{C}}(B, Z) \xrightarrow{\simeq} \operatorname{Map}_{\mathcal{C}}(A, Z).$$

Definition 30.2 (Local equivalence)**.** A map $g\colon X \to Y$ is an *S-local equivalence* if for every S-local object Z,

$$g^*\colon \operatorname{Map}_{\mathcal{C}}(Y, Z) \xrightarrow{\simeq} \operatorname{Map}_{\mathcal{C}}(X, Z)$$

is an equivalence.

So local objects cannot tell the maps in S from equivalences, and the local equivalences are exactly the maps that no local object can tell from an equivalence. This class contains S and is usually much bigger.

Definition 30.3 (Localization functor)**.** An *S-localization* is a functor $L\colon \mathcal{C} \to \mathcal{C}$ with a natural map $\eta_X\colon X \to LX$ such that, for every X, the object LX is S-local and η_X is an S-local equivalence.

Then

$$\mathrm{Map}_{\mathcal{C}}(LX, Z) \simeq \mathrm{Map}_{\mathcal{C}}(X, Z) \qquad (Z \text{ local}),$$

so LX is the universal local object under X, and $L^2 \simeq L$. Said differently, L is left adjoint to the inclusion of the full subcategory of local objects:

$$\mathcal{C} \underset{i}{\overset{L}{\rightleftarrows}} \mathcal{C}^{\mathrm{loc}} \qquad L \dashv i.$$

If $\mathcal{C}$ is presentable and S is a set, the localization exists and $\mathcal{C}^{\mathrm{loc}}$ is again presentable.

30.3 Basic examples

In abelian groups, take S to be the single map $p\colon \mathbb{Z} \to \mathbb{Z}$. A group A is local when multiplication by p on A is an isomorphism, so the local objects are the $\mathbb{Z}[1/p]$-modules and the localization is $A \mapsto A[1/p]$. To get rational vector spaces one has to invert every prime.

In commutative monoids, localizing at

$$\mathbb{N} \longrightarrow \mathbb{Z}$$

gives group completion: the local objects are the abelian groups. The same map, read in E_∞-spaces, gives the group completion that appeared on Days 2 and 3.

In spaces, localizing at $S^{n+1} \to *$ gives the n-truncated spaces, since Z is n-truncated exactly when

$$\mathrm{Map}(*, Z) \xrightarrow{\simeq} \mathrm{Map}(S^{n+1}, Z)$$

is an equivalence. The localization functor is the Postnikov truncation $\tau_{\le n}$. At the silly extreme, localizing at $\varnothing \to *$ leaves only contractible spaces.

Sullivan's localizations of nilpotent spaces are also of this form: one inverts degree maps on spheres. Rationalization inverts the degree-p maps for all primes p, and p-localization inverts them for the primes different from p.

Theorem 30.4 (Sullivan rational homotopy theory)**.** *The rational homotopy category of simply connected spaces of finite rational type is contravariantly equivalent to the homotopy category of 1-connected commutative differential graded $\mathbb{Q}$-algebras of finite type. Here 1-connected means $H^0 \cong \mathbb{Q}$ and $H^1 = 0$. The functor from spaces is the polynomial de Rham algebra A_{PL}.*

Rationalization does what one expects on homotopy groups: for simply connected X,

$$\pi_n(X_{\mathbb{Q}}) \cong \pi_n(X) \otimes \mathbb{Q} \qquad (n \ge 2).$$

In spectra, rationalization is localization at $H\mathbb{Q}$, and it is given by a smash product,

$$L_{H\mathbb{Q}} X \simeq H\mathbb{Q} \wedge X.$$

30.4 The small-object construction

How is $L_S X$ built? In a simplicial model category, first replace the maps in $S = \{A_\alpha \to B_\alpha\}$ by cofibrations between cofibrant objects. For Z to be local, every map $A_\alpha \to Z$ must extend over B_α. Given a map $A_\alpha \to X$ that does not extend, we can force an extension by forming the pushout

$$\begin{array}{ccc} A_\alpha & \longrightarrow & X \\ \downarrow & & \downarrow \\ B_\alpha & \longrightarrow & X \cup_{A_\alpha} B_\alpha. \end{array}$$

This handles π_0 of the mapping spaces. Locality is a condition on the whole mapping spaces, so for each $n \geq 0$ we also attach along the pushout-products

$$(A_\alpha \otimes \Delta^n) \underset{A_\alpha \otimes \partial\Delta^n}{\cup} (B_\alpha \otimes \partial\Delta^n) \longrightarrow B_\alpha \otimes \Delta^n.$$

The case $n = 1$, for instance, makes any two extensions homotopic: the domain is two copies of B_α glued along a cylinder on A_α, and for local Z

$$\mathrm{Map}(B_\alpha \cup^h_{A_\alpha} B_\alpha, Z) \simeq \mathrm{Map}(B_\alpha, Z) \times^h_{\mathrm{Map}(A_\alpha, Z)} \mathrm{Map}(B_\alpha, Z).$$

Now run the small-object argument with all of these maps, together with the generating acyclic cofibrations. The result is a natural map

$$X \longrightarrow L_S X$$

whose target is fibrant and local.

30.5 Left Bousfield localization of a model category

In model-category terms, localization keeps the category and changes the model structure. The *left Bousfield localization* $L_S\mathcal{M}$ of a model category $\mathcal{M}$, when it exists, has

$$\begin{aligned} \text{cofibrations in } L_S\mathcal{M} &= \text{cofibrations in } \mathcal{M}, \\ \text{weak equivalences in } L_S\mathcal{M} &= S\text{-local equivalences}, \end{aligned}$$

and the fibrations are determined by lifting. There are more weak equivalences than before and so fewer fibrations. The fibrant objects are the fibrant objects of $\mathcal{M}$ that are S-local, and fibrant replacement in $L_S\mathcal{M}$ is the localization functor. The identity functors form a Quillen adjunction

$$\mathcal{M} \underset{\text{id}}{\overset{\text{id}}{\rightleftarrows}} L_S\mathcal{M}$$

with the left adjoint going from $\mathcal{M}$ to $L_S\mathcal{M}$.

31 Existence and Monoidal Localization

31.1 Combinatorial model categories

Definition 31.1 (Combinatorial model category)**.** A model category is *combinatorial* if its underlying category is locally presentable and its model structure is cofibrantly generated.

Existence comes from a recognition theorem of Jeff Smith.

Theorem 31.2 (Smith recognition)**.** *Let $\mathcal{C}$ be locally presentable, let I be a set of maps, and let W be an accessible, accessibly embedded class of maps. Assume:*

1. *W has two-out-of-three and is closed under retracts;*
2. *I-inj $\subseteq W$;*
3. *$W \cap \mathrm{cof}(I)$ is closed under pushouts and transfinite composition.*

Then $\mathcal{C}$ has a combinatorial model structure with weak equivalences W and cofibrations $\mathrm{cof}(I)$.

The theorem needs no generating acyclic cofibrations, which makes it usable here: for a left proper combinatorial model category $\mathcal{M}$ and a set S, taking W to be the S-local equivalences shows that $L_S\mathcal{M}$ exists.

The ∞-categorical statement is cleaner. A presentable ∞-category can be localized at any set of maps S, and the local equivalences are the strongly saturated class generated by S. That is the smallest class containing S and the equivalences that has two-out-of-three and is closed under pushouts and under colimits in the arrow category.

31.2 Stable localizations

Let $\mathcal{C}$ be a stable ∞-category and L a localization of it. The following are equivalent:

1. the subcategory of local objects is stable;
2. L is exact;
3. the L-acyclic objects, those X with $LX \simeq 0$, form a stable subcategory;
4. the local equivalences are closed under suspension and desuspension.

In that case the fiber of $X \to LX$ defines the acyclization functor Γ, and there is a natural fiber sequence

$$\Gamma X \longrightarrow X \longrightarrow LX$$

with ΓX acyclic and LX local.

Localization at a spectrum E is stable, because $E \wedge -$ is exact. The truncation $\tau_{\leq n}$ on spectra is a localization that is not stable: its local objects are closed under desuspension but not under suspension.

31.3 Localization and completion at a prime

There are three operations on spectra associated with a prime p, and they are easy to confuse.

- *p-localization* inverts every prime other than p: $X_{(p)} \simeq X \wedge \mathbb{S}_{(p)}$.
- *Inverting p* inverts only p: $X[1/p] \simeq X \wedge \mathbb{S}[1/p]$.
- *p-completion* keeps what mod-p homology detects.

The first two are given by smashing with a localized sphere. Completion is

$$X^\wedge_p \simeq \operatorname*{holim}_r X/p^r, \qquad X/p^r = \mathrm{cof}\big(p^r \colon X \to X\big),$$

which is localization at the mod-p Moore spectrum $\mathbb{S}/p$. For bounded-below X it agrees with localization at $H\mathbb{F}_p$, and in general the two can differ. Completion is not given by smashing with anything, since it does not commute with infinite wedges.

31.4 Smashing localizations

Let $\mathcal{C}$ be a presentably symmetric monoidal stable ∞-category.

Definition 31.3 (Smashing localization)**.** A localization L of $\mathcal{C}$ is *smashing* if there is a natural equivalence

$$LX \simeq X \otimes L\mathbf{1}$$

for every X.

So a smashing localization is determined by the single object $L\mathbf{1}$, which is an idempotent commutative algebra:

$$L\mathbf{1} \otimes L\mathbf{1} \simeq L\mathbf{1}.$$

Conversely, tensoring with any idempotent commutative algebra is a smashing localization. Rationalization, p-localization, and finite localizations are smashing. Postnikov truncation is not, and neither is p-completion.

31.5 The tensor product after localization

A localization that is not smashing can still be compatible with the tensor product. Suppose that $f \otimes X$ is a local equivalence whenever f is one. Then the local subcategory gets a tensor product

$$X \otimes_{\mathrm{loc}} Y := L(iX \otimes iY), \qquad \mathbf{1}_{\mathrm{loc}} = L\mathbf{1},$$

for which L is symmetric monoidal and the inclusion i is lax symmetric monoidal. In particular L sends commutative algebras to commutative algebras. This is the situation for $K(n)$-local spectra, where the product is $L_{K(n)}(X \wedge Y)$.

32 Local Duality and Completion

32.1 Torsion, local, and complete modules

Let R be an E_∞-ring and $I \subseteq \pi_0 R$ a finitely generated ideal. Write R/I for the Koszul object: choose generators $x_1, \dots, x_k$ of I and take the tensor product of the cofibers of the maps $x_i \colon R \to R$.

Definition 32.1 (Torsion, local, and complete)**.** An R-module M is

1. *I-torsion* if it lies in the localizing subcategory generated by R/I;
2. *I-local* if $\mathrm{Map}_R(T, M) \simeq *$ for every I-torsion module T;
3. *I-complete* if $\mathrm{Map}_R(N, M) \simeq *$ for every I-local module N.

Let Γ_I be the right adjoint to the inclusion of the torsion modules, and L_I the left adjoint to the inclusion of the local modules. They fit into a fiber sequence

$$\Gamma_I M \longrightarrow M \longrightarrow L_I M, \tag{5.1}$$

which is the sequence $\Gamma X \to X \to LX$ from before. The completion functor Λ^I is left adjoint to the inclusion of the complete modules, and local duality is the adjunction

$$\mathrm{Map}_R(\Gamma_I M, N) \simeq \mathrm{Map}_R(M, \Lambda^I N). \tag{5.2}$$

A consequence is that Γ_I and Λ^I restrict to inverse equivalences between the torsion modules and the complete modules.

For a Noetherian ring this is the familiar derived completion,

$$\Lambda^I M \simeq \operatorname*{holim}_r (M \otimes_R R/I^r), \tag{5.3}$$

and in general Λ^I is Bousfield localization at R/I in Mod_R. With $R = \mathbb{S}$ and $I = (p)$ this is p-completion.

When $I = (x)$ is principal, the local part is

$$L_I M \simeq M \otimes_R R[x^{-1}].$$

For an ideal with several generators there is no single ring to invert into, and $L_I M$ is computed by the Čech complex on the localizations $R[x_i^{-1}]$.

32.2 Adams towers and cobar resolutions

Completion can also be taken with respect to an algebra. Let A be an E_1-algebra in a stable symmetric monoidal ∞-category, with unit $\mathbf{1} \to A$, and let

$$J = \mathrm{fib}(\mathbf{1} \longrightarrow A).$$

Tensor powers of J give the Adams tower

$$\cdots \longrightarrow J^{\otimes 3} \otimes X \longrightarrow J^{\otimes 2} \otimes X \longrightarrow J \otimes X \longrightarrow X.$$

The cofibers

$$C_n X = \mathrm{cof}\big(J^{\otimes (n+1)} \otimes X \longrightarrow X\big)$$

are built from finitely many A-modules, and the A-completion of X is their limit,

$$X^\wedge_A := \lim_n C_n X.$$

The same thing can be written cosimplicially. The cobar construction is

$$X \longrightarrow A \otimes X \;\substack{\longrightarrow\\ \longrightarrow}\; A^{\otimes 2} \otimes X \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; A^{\otimes 3} \otimes X \longrightarrow \cdots,$$

and its totalization is the completion:

$$X^\wedge_A \simeq \mathrm{Tot}\big(A^{\otimes (\bullet+1)} \otimes X\big), \qquad X \longrightarrow X^\wedge_A. \tag{5.4}$$

The spectral sequence of this tower is the A-based Adams spectral sequence, and it converges to $\pi_* X^\wedge_A$. If A is descendable, then $X \to X^\wedge_A$ is an equivalence for every X.

For $A = H\mathbb{F}_p$ and X a bounded-below spectrum, $X^\wedge_A$ is the p-completion, and we get the classical Adams spectral sequence.

32.3 Geometry of local duality

All of this has a geometric meaning. Let $X = \mathrm{Spec}\, R$, $Z = V(I)$, and $U = X \setminus Z$. Then

algebra	geometry
I-torsion	quasi-coherent sheaves supported on Z
I-local	quasi-coherent sheaves on U
I-complete	quasi-coherent sheaves on $\widehat{X}_Z$,

where $\widehat{X}_Z$ is the formal completion of X along Z. A sheaf on X can be recovered from its restriction to U, its completion along Z, and gluing data over the punctured formal

neighborhood $\widehat{X}^{\circ}_{Z}$. This is formal gluing: the square

$$\begin{array}{ccc} \mathrm{QCoh}(X) & \longrightarrow & \mathrm{QCoh}(U) \\ \downarrow & & \downarrow \\ \mathrm{QCoh}(\widehat{X}_Z) & \longrightarrow & \mathrm{QCoh}(\widehat{X}^{\circ}_{Z}) \end{array} \tag{5.5}$$

is a pullback. The arithmetic fracture square is the case $R = \mathbb{S}$, $I = (p)$.

33 Chromatic Localization

Fix a prime p. In this section all spectra are p-local.

33.1 Morava K-theories and height

For each $0 < n < \infty$ there is a Morava K-theory $K(n)$, with

$$K(n)_* \cong \mathbb{F}_p[v_n^{\pm 1}], \qquad |v_n| = 2(p^n - 1). \tag{5.6}$$

By convention

$$K(0) = H\mathbb{Q}, \qquad K(\infty) = H\mathbb{F}_p.$$

The coefficient ring is a graded field, so every $K(n)$-module is a wedge of suspensions of $K(n)$, and $K(n)$-homology has a Künneth isomorphism.

A nonzero finite spectrum F has *type* n if

$$K(m)_*F = 0 \ \text{ for } m < n \qquad \text{and} \qquad K(n)_*F \neq 0.$$

33.2 Nilpotence and thick subcategories

The Morava K-theories together detect nilpotence. Let $f\colon F \to X$ be a map from a finite spectrum. If $K(n)_*f = 0$ for all $0 \le n \le \infty$, then

$$f^{\wedge N} \simeq 0$$

for some N. The ring-spectrum form of the nilpotence theorem says that an element in the homotopy of a ring spectrum is nilpotent if its image in MU-homology is.

Let $\mathcal{C}_0$ be the category of all finite spectra, and for $n \ge 1$ let

$$\mathcal{C}_n = \{F \in \mathbf{Sp}^{\mathrm{fin}}_{(p)} : K(n-1)_*F = 0\}.$$

If $K(n-1)_*F = 0$ then also $K(m)_*F = 0$ for all $m < n$, so $\mathcal{C}_n$ consists of the spectra of type at least n together with zero. Mitchell showed that there are finite spectra of every type, so the inclusions are strict:

$$\mathcal{C}_0 \supsetneq \mathcal{C}_1 \supsetneq \mathcal{C}_2 \supsetneq \cdots. \tag{5.7}$$

Theorem 33.1 (Hopkins–Smith)**.** *Every nonzero thick subcategory of finite p-local spectra is one of the $\mathcal{C}_n$.*

33.3 Periodic self-maps and telescopes

Let F have type n. A *v_n-self-map* of F is a map

$$v\colon \Sigma^d F \longrightarrow F$$

that induces an isomorphism on $K(n)_*F$ and a nilpotent map on $K(m)_*F$ for $m \neq n$. The periodicity theorem says that every type-n spectrum has one, and that any two of them have iterates that agree.

Inverting v gives the telescope

$$T(n) = \operatorname{colim}\bigl(F \xrightarrow{v} \Sigma^{-d}F \xrightarrow{v} \Sigma^{-2d}F \longrightarrow \cdots\bigr). \tag{5.8}$$

The spectrum $T(n)$ depends on F and v, but localization at $T(n)$ does not. There are now two candidates for "height exactly n," and the telescope conjecture asked whether they agree:

$$L_{T(n)} \overset{?}{\simeq} L_{K(n)}. \tag{5.9}$$

This is true for $n \leq 1$ and false for all $n \geq 2$.

33.4 The chromatic tower

Let

$$L_n = L_{K(0)\vee K(1)\vee\cdots\vee K(n)}.$$

This is the same as localization at the Johnson–Wilson theory $E(n)$, and it is smashing. Since L_n inverts fewer maps than L_{n-1}, there are natural maps $L_nX \to L_{n-1}X$, which form the chromatic tower

$$\cdots \longrightarrow L_nX \longrightarrow L_{n-1}X \longrightarrow \cdots \longrightarrow L_0X.$$

The bottom is the rationalization L_0X. The fibers

$$M_nX = \operatorname{fib}(L_nX \longrightarrow L_{n-1}X)$$

are the monochromatic layers.

Theorem 33.2 (Chromatic convergence)**.** *For a finite p-local spectrum F, the natural map*

$$F \longrightarrow \operatorname*{holim}_n L_nF$$

is an equivalence.

The layer M_nX and the localization $L_{K(n)}X$ determine each other, so the building blocks are the $K(n)$-local spectra.

33.5 The $K(n)$-local sphere

The $K(n)$-localization can be computed from L_n by a completion. There are generalized Moore spectra M_I with

$$BP_*M_I \cong BP_*/I, \qquad I = (p^{i_0}, v_1^{i_1}, \ldots, v_{n-1}^{i_{n-1}}),$$

for a cofinal system of exponents, and

$$L_{K(n)}X \simeq \operatorname*{holim}_I (L_n X \wedge M_I); \tag{5.10}$$

see [HS99]. This is the chromatic analogue of $X^\wedge_p = \operatorname{holim}_r X/p^r$.

The other description is by descent. Morava E-theory E_n has

$$(E_n)_0 \cong W(\mathbb{F}_{p^n})[[u_1, \dots, u_{n-1}]], \qquad (E_n)_* \cong (E_n)_0[u^{\pm 1}], \quad |u| = -2,$$

and the extended Morava stabilizer group $\mathbb{G}_n$ acts on it. The Devinatz–Hopkins theorem says that

$$L_{K(n)}\mathbb{S} \simeq E_n^{h\mathbb{G}_n}, \tag{5.11}$$

where the fixed points are continuous ones for the profinite group $\mathbb{G}_n$. For a finite spectrum X there is a homotopy fixed point spectral sequence

$$H^s_c(\mathbb{G}_n; \pi_t(E_n \wedge X)) \Longrightarrow \pi_{t-s} L_{K(n)} X. \tag{5.12}$$

At height 1, E_1 is p-adic K-theory and $\mathbb{G}_1 = \mathbb{Z}_p^\times$ acts by Adams operations.

34 Spectral Algebraic Geometry

34.1 Flat maps and fpqc descent

Spectral algebraic geometry is algebraic geometry with E_∞-rings in place of commutative rings. The first thing to define is flatness.

Definition 34.1 (Flat map)**.** A map $f\colon A \to B$ of E_∞-rings is *flat* if $\pi_0 B$ is flat over $\pi_0 A$ and the natural maps

$$\pi_n A \otimes_{\pi_0 A} \pi_0 B \longrightarrow \pi_n B$$

are isomorphisms for all n. It is *faithfully flat* if in addition $\pi_0 A \to \pi_0 B$ is faithfully flat.

So a flat map is flat on π_0 and the higher homotopy groups of B are obtained by base change. The fpqc topology is the one generated by faithfully flat quasi-compact covers. A faithfully flat map has an Amitsur complex, the cosimplicial ring

$$B \rightrightarrows B \otimes_A B \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; B \otimes_A B \otimes_A B \longrightarrow \cdots,$$

and faithfully flat descent says that A is recovered from it,

$$A \simeq \operatorname{Tot}\big(B^{\otimes_A (\bullet+1)}\big), \tag{5.13}$$

and so is its category of modules:

$$\operatorname{Mod}_A \simeq \operatorname{Tot}\big([q] \longmapsto \operatorname{Mod}_{B^{\otimes_A (q+1)}}\big). \tag{5.14}$$

This is the cobar construction (5.4) again, in a case where the completion map is an equivalence.

34.2 Spectral affine schemes and stacks

The category of affine spectral schemes is

$$\mathrm{Aff}^{\mathrm{sp}} = \mathrm{CAlg}(\mathbf{Sp})^{\mathrm{op}},$$

and we think of $\operatorname{Spec} A$ through its functor of points

$$\operatorname{Spec} A(B) = \mathrm{Map}_{\mathrm{CAlg}(\mathbf{Sp})}(A, B). \tag{5.15}$$

Definition 34.2 (Spectral stack)**.** A *spectral stack* (for the fpqc topology) is a functor

$$\mathcal{X} \colon \mathrm{CAlg}(\mathbf{Sp}) \longrightarrow \mathcal{S}$$

that satisfies fpqc descent.

Quasi-coherent sheaves on an affine are modules,

$$\mathrm{QCoh}(\operatorname{Spec} A) \simeq \mathrm{Mod}_A, \tag{5.16}$$

with global sections of the structure sheaf giving back A. On a general stack they are defined by descent, as compatible families of modules over all affines mapping to $\mathcal{X}$:

$$\mathrm{QCoh}(\mathcal{X}) \simeq \lim_{\operatorname{Spec} A \to \mathcal{X}} \mathrm{Mod}_A\,. \tag{5.17}$$

The structure sheaf $\mathcal{O}_{\mathcal{X}}$ has an E_∞-ring of global sections $\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$.

A connective spectral scheme has an underlying classical scheme,

$$t_0(\operatorname{Spec} A) = \operatorname{Spec} \pi_0 A,$$

and the higher homotopy groups $\pi_i A$ are quasi-coherent sheaves on it [BGH22]. So a spectral scheme is a classical scheme with extra homotopical structure on its sheaf of functions.

34.3 Spherical monoid algebras

The simplest examples come from commutative monoids. For a commutative monoid M, the spherical monoid algebra is

$$\mathbb{S}[M] = \Sigma^\infty_+ M,$$

and the spectral affine line and multiplicative group are

$$\mathbb{A}^1_{\mathbb{S}} = \operatorname{Spec} \mathbb{S}[\mathbb{N}], \qquad \mathbb{G}_{m,\mathbb{S}} = \operatorname{Spec} \mathbb{S}[\mathbb{Z}]. \tag{5.18}$$

The inclusion $\mathbb{N} \to \mathbb{Z}$ gives the open embedding $\mathbb{G}_{m,\mathbb{S}} \to \mathbb{A}^1_{\mathbb{S}}$. Base change along $\mathbb{S} \to H\mathbb{Z}$ gives $H\mathbb{Z}[t]$ and $H\mathbb{Z}[t^{\pm 1}]$, the ordinary affine line and multiplicative group.

34.4 Formal spectra and formal groups

For an ideal $I \subseteq \pi_0 A$, the formal spectrum is the colimit of the infinitesimal neighborhoods of $V(I)$,

$$\operatorname{Spf}(A, I) \simeq \operatorname*{colim}_r \operatorname{Spec}(A/I^r), \tag{5.19}$$

and its quasi-coherent sheaves are the I-complete A-modules, as in the table in the section on local duality.

The formal objects we care about are one-dimensional formal groups. Their moduli stack is $\mathcal{M}_{FG}$, whose value on an E_∞-ring A is the ∞-groupoid of formal groups over A,

$$\mathcal{M}_{FG}(A) \simeq \mathrm{FGroup}(A)^{\simeq}.$$

Classically a formal group with a chosen coordinate is a formal group law, and an isomorphism is a change of coordinate

$$x' = a_1 x + a_2 x^2 + \cdots, \qquad a_1 \in (\pi_0 A)^\times.$$

The isomorphism is called strict when $a_1 = 1$.

34.5 Preorientations and orientations

A complex oriented ring spectrum R already has a formal group, namely $\operatorname{Spf} R^0(\mathbb{CP}^\infty)$. An orientation of a formal group over R is what identifies it with this one. Let G be a one-dimensional formal group over an E_∞-ring R, which need not be connective. A *preorientation* of G is a based map

$$e\colon S^2 \longrightarrow G(\tau_{\geq 0} R).$$

Let ω_G be the dualizing line of G, an invertible R-module playing the role of the cotangent space at the identity. A preorientation determines a Bott map

$$\beta_e\colon \omega_G \longrightarrow \Sigma^{-2} R. \tag{5.20}$$

Definition 34.3 (Orientation)**.** A preorientation e is an *orientation* if its Bott map β_e is an equivalence.

An orientation forces R to be weakly 2-periodic; see [BGH22].

34.6 Periodic complex bordism and descent

The universal example is periodic complex bordism MUP from Day 4. Its orientation gives an fpqc cover

$$\operatorname{Spec} MUP \longrightarrow \mathcal{M}_{FG}^{\mathrm{or}}$$

of the moduli stack of oriented formal groups. The Čech nerve of the cover has terms

$$\operatorname{Spec}\bigl(MUP^{\otimes(q+1)}\bigr),$$

so global functions are computed by the Amitsur complex of $\mathbb{S} \to MUP$:

$$\Gamma(\mathcal{M}_{FG}^{\mathrm{or}}, \mathcal{O}) \simeq \operatorname{Tot}\bigl(MUP^{\otimes(\bullet+1)}\bigr) \simeq \mathbb{S}. \tag{5.21}$$

The sphere spectrum is the ring of functions on the moduli of oriented formal groups, and the descent spectral sequence is the periodic Adams–Novikov spectral sequence.

34.7 Elliptic curves and topological modular forms

An elliptic curve has a formal group, its completion at the identity section. This gives a map of moduli stacks

$$\mathcal{M}_{Ell} \longrightarrow \mathcal{M}_{FG},$$

and we can ask for orientations of the formal group of an elliptic curve:

$$\begin{array}{ccc} \mathcal{M}_{Ell}^{\mathrm{or}} & \longrightarrow & \mathcal{M}_{FG}^{\mathrm{or}} \\ \downarrow & & \downarrow \\ \mathcal{M}_{Ell} & \longrightarrow & \mathcal{M}_{FG}. \end{array} \tag{5.22}$$

The square is a pullback, and $\mathcal{M}_{Ell}^{\mathrm{or}}$ can be described as the classical stack $\mathcal{M}_{Ell}$ with a sheaf $\mathcal{O}^{\mathrm{top}}$ of E_∞-rings on it (the Goerss–Hopkins–Miller theorem). On an affine étale chart $\operatorname{Spec} R_0 \to \mathcal{M}_{Ell}$ its value is an elliptic cohomology theory R, which is even and weakly 2-periodic:

$$\pi_{2k+1}R = 0, \qquad \pi_{2k}R \cong (\pi_2 R)^{\otimes_{\pi_0 R} k},$$

with $\pi_0 R = R_0$ and negative powers meaning powers of the inverse line. Global sections are the periodic topological modular forms,

$$\Gamma(\mathcal{M}_{Ell}^{\mathrm{or}}, \mathcal{O}) \simeq \Gamma(\mathcal{M}_{Ell}, \mathcal{O}^{\mathrm{top}}) \simeq TMF. \tag{5.23}$$

There are three versions: TMF is the periodic one above, Tmf is the global sections over the compactified moduli stack, and tmf is the connective cover of Tmf.

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